# Centre of Mass

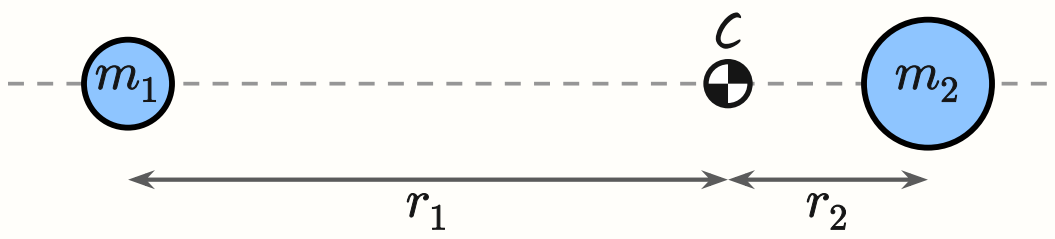
Class 11 · point masses and rigid bodies · UNIFORM $g$ · SMOOTH surfaces unless stated · $M$ = TOTAL mass of the system

## 1 Definition

$$\vec{r}_{cm} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + \dots}{m_1 + m_2 + \dots} = \frac{\sum m_i\vec{r}_i}{M}$$

$m_i$ = mass of particle $i$ (kg) · $\vec{r}_i$ = its position vector (m) · $M = \sum m_i$ = TOTAL mass · the mass-weighted AVERAGE position

## 2 Two particles




$$m_1 r_1 = m_2 r_2 \qquad r_1 = \frac{m_2\, r}{M}, \quad r_2 = \frac{m_1\, r}{M}$$

$r = r_1 + r_2$ = separation · C lies ON the line joining them, CLOSER to the HEAVIER mass · $m_1 = 2\,\text{kg}$, $m_2 = 6\,\text{kg}$, $r = 4\,\text{m}$: $r_1 = 3\,\text{m}$, $r_2 = 1\,\text{m}$

- C may lie OUTSIDE the material (ring, L-shaped rod, boomerang). For a UNIFORM symmetric body it lies on every axis of symmetry.
- Centre of GRAVITY coincides with C only in a UNIFORM field $g$ (true for bodies small compared with the Earth).

# Syllabus Checklist

JEE Main + JEE Advanced + NEET syllabus + standard results

## 1 JEE Main + JEE Advanced + NEET syllabus


- ✓ Centre of mass of a system of particles — p1, p3
- ✓ Rod, ring, arc, semicircular disc, sector — p4–p6
- ✓ Hemisphere (solid, shell), cone, triangle — p7, p8
- ✓ Composite bodies; cavity (removed part) method — p8, p9
- ✓ Motion of C: $\vec{v}_{cm}$, $\vec{a}_{cm}$, $\vec{F}_{ext} = M\vec{a}_{cm}$ — p10, p11
- ✓ Linear momentum, impulse, $F$–$t$ graph — p12, p13
- ✓ Conservation of linear momentum (condition) — p14
- ✓ Variable mass: rocket, conveyor belt, chain — p15, p16
- ✓ Elastic, inelastic, perfectly inelastic collisions — p17
- ✓ Coefficient of restitution $e$ — p18
- ✓ 1-D elastic collision and special cases — p19, p20
- ✓ Collisions with $e$; KE loss $\frac{1}{2}\mu v_{rel}^2(1-e^2)$ — p21
- ✓ Ballistic pendulum — p22
- ✓ Oblique (2-D) collisions; equal masses at $90^\circ$ — p23, p24
- ✓ Collision with a wall; bouncing ball — p25, p26
- ✓ Spring–block two-body systems, C frame — p27, p28
- ✓ Explosions; man on boat; wedge–block — p29–p31


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## 2 Also covered


- ✓ Standard results; common traps — p32, p33
- ✓ Solved questions (8 Main + 6 Advanced level) — p34–p38

# Centre of Mass: Particles

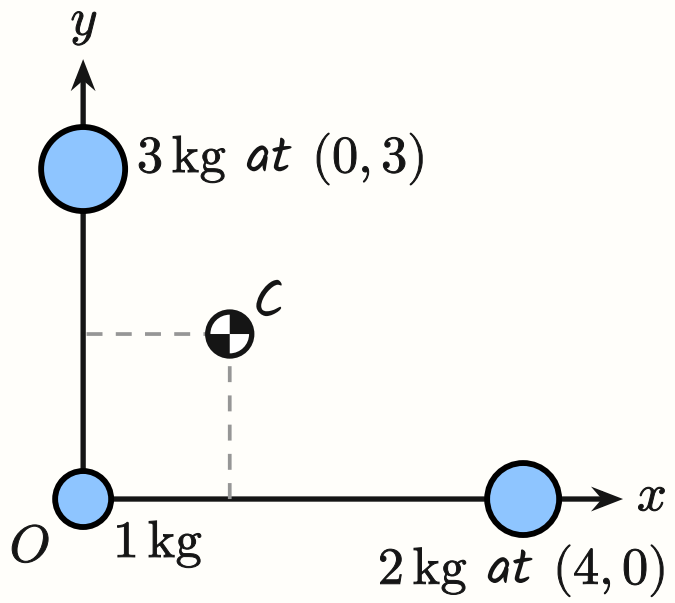
*point masses · any origin (C itself does not depend on the choice)*

## 1 Component form

$$x_{cm} = \frac{\sum m_i x_i}{M}, \quad y_{cm} = \frac{\sum m_i y_i}{M}, \quad z_{cm} = \frac{\sum m_i z_i}{M}$$

$(x_i, y_i, z_i)$ = coordinates of particle $i$ (m) · a group of particles can be replaced by its total mass placed at ITS own centre of mass

## 2 Example: three particles




**Example**

1 kg at $(0, 0)$, 2 kg at $(4, 0)$, 3 kg at $(0, 3)$ (m):

$x_{cm} = \frac{2\times4}{6} = 1.33\,\text{m}$

$y_{cm} = \frac{3\times3}{6} = 1.5\,\text{m}$

- Equal masses at the corners of a square, a regular polygon or a cube: C is at the geometric centre.
- Adding a mass anywhere except at C moves C TOWARDS it; C always lies inside the smallest convex region containing all the particles.

# Centre of Mass: Continuous Bodies

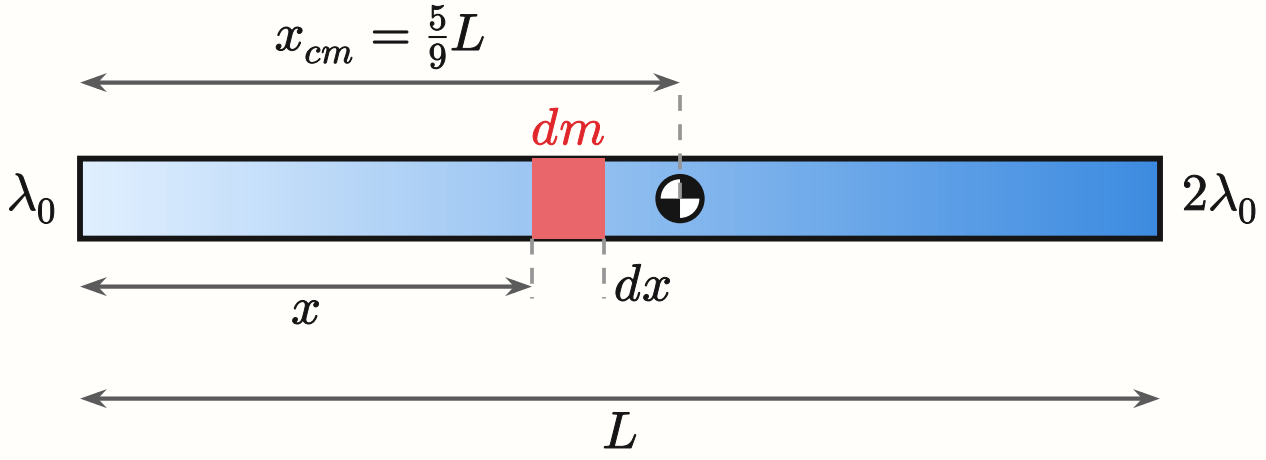
*$x$ from the left end · $\lambda$ = mass per unit length*

## 1 Integral form

$$x_{cm} = \frac{\int x\,dm}{\int dm} \qquad dm = \lambda\,dx \ \ (\text{rod})$$

*$dm$ = mass of a small element at $x$ · lamina: $dm = \sigma\,dA$ (mass per unit area), solid: $dm = \rho\,dV$ (density) · limits cover the WHOLE body*

## 2 Rods




*uniform:* $\dfrac{L}{2}$ $\qquad$ $\lambda \propto x$ : $\dfrac{2L}{3}$

$$\lambda = \lambda_0\left(1 + \frac{x}{L}\right) : \ x_{cm} = \frac{5L}{9}$$

*shading: $\lambda$ grows from $\lambda_0$ to $2\lambda_0$; $\int_0^L x\lambda\,dx = \frac{5}{6}\lambda_0 L^2$, $\int_0^L \lambda\,dx = \frac{3}{2}\lambda_0 L$ · $C$ shifts towards the DENSER end*

**Example**

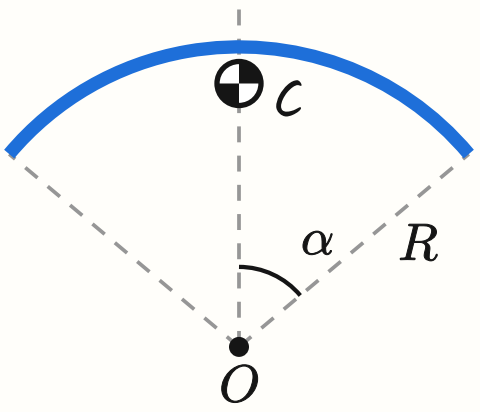
*Rod of length $L = 1.8\,\text{m}$ with $\lambda = \lambda_0\,(1 + \frac{x}{L})$:*

$x_{cm} = \frac{5}{9} \times 1.8 = 1.0\,\text{m}$ *from the lighter end.*

# Arcs and Rings

**UNIFORM thin wire of radius $R$ · $O$ = centre of the circle · $\alpha$ in RADIANS**

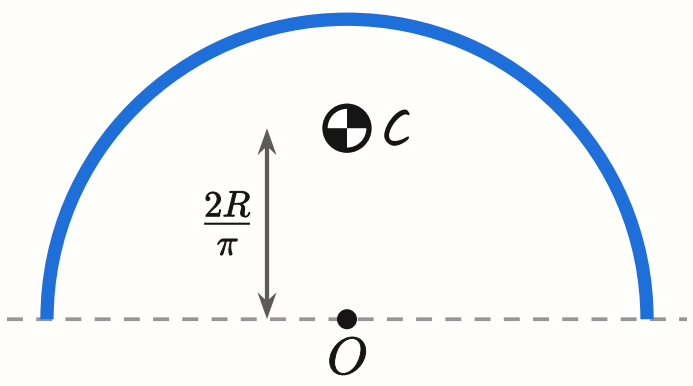
## 1 Arc of angle $2\alpha$




$$OC = \frac{R\sin\alpha}{\alpha}$$

*on the bisector of the arc · $\alpha$ = HALF the angle subtended at $O$, in rad · from $dm = \frac{M}{2\alpha}d\theta$ at height $R\cos\theta$, $\theta$ from $-\alpha$ to $\alpha$*

## 2 Semicircular ring and quarter ring




*semicircle:* $OC = \frac{2R}{\pi}$

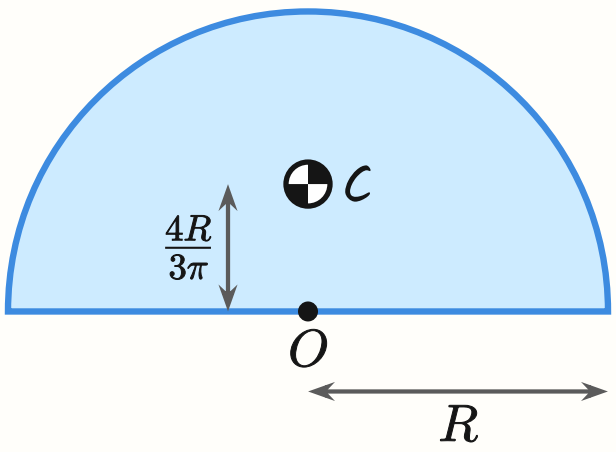
*$\alpha = \frac{\pi}{2}$, measured from the diameter · QUARTER ring ($\alpha = \frac{\pi}{4}$): $OC = \frac{2\sqrt{2}R}{\pi}$ on the bisector, i.e. $\left(\frac{2R}{\pi}, \frac{2R}{\pi}\right)$ from $O$ along the two radii*

- *Full ring: C at the centre $O$, where there is NO mass. C of any arc lies INSIDE the circle, never on the wire (except $\alpha \to 0$).*
- *Semicircular ring of $R = 22\,\text{cm}$: C is $\frac{2\times 22}{\pi} = 14.0\,\text{cm}$ from the centre, on the axis of symmetry.*

# Discs and Sectors

**UNIFORM lamina (plate) of radius $R$ · $O$ = centre · $\alpha$ in RADIANS**

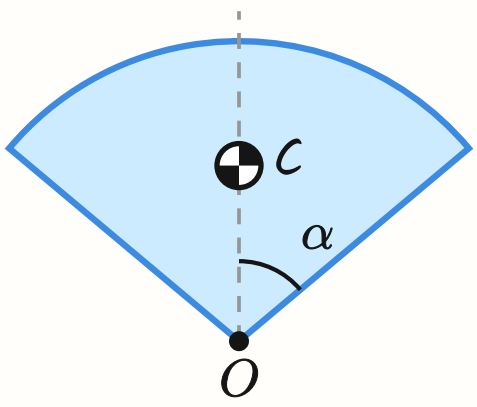
## 1 Semicircular disc




$$OC = \frac{4R}{3\pi}$$

*from the straight edge, on the axis of symmetry · smaller than the ring value $\frac{2R}{\pi}$ (mass nearer $O$)*

## 2 Sector of angle $2\alpha$; quarter disc




$$OC = \frac{2R\sin\alpha}{3\alpha}$$

*$\alpha = \frac{\pi}{2}$: semicircular disc $\frac{4R}{3\pi}$ · quarter disc: $\left(\frac{4R}{3\pi}, \frac{4R}{3\pi}\right)$ from the corner, along both edges*

- *Why $\frac{2}{3}$: a sector is made of thin triangles from $O$; each has its C at $\frac{2}{3}R$ from $O$, so the sector behaves like an ARC of radius $\frac{2}{3}R$: $OC = \frac{2}{3}R\,\frac{\sin\alpha}{\alpha}$.*

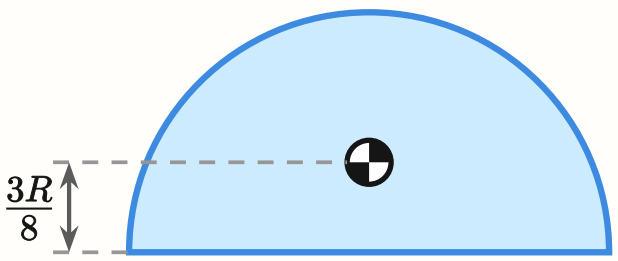
**Example**

*Semicircular plate of radius $R = 30\,\text{cm}$: $OC = \frac{4\times30}{3\pi} = 12.7\,\text{cm}$; a semicircular RING of the same radius: $\frac{2\times30}{\pi} = 19.1\,\text{cm}$.*

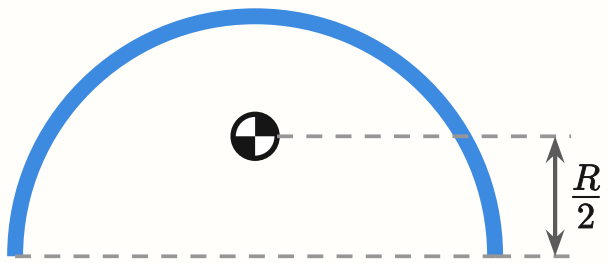
# Hemisphere and Cone

UNIFORM bodies • distances from the FLAT face (base) along the axis

## 1 Hemispheres



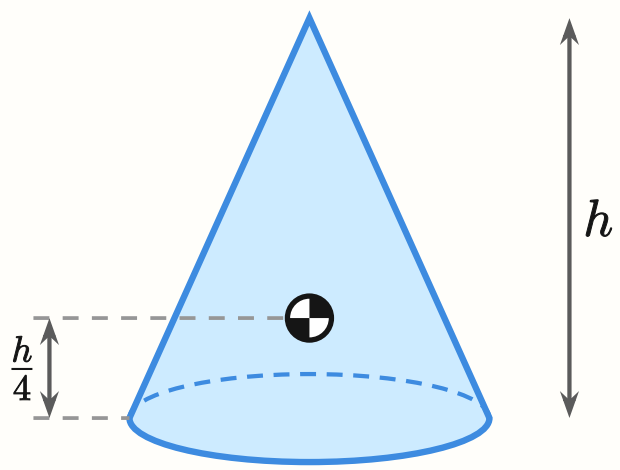

SOLID: $\frac{3R}{8}$



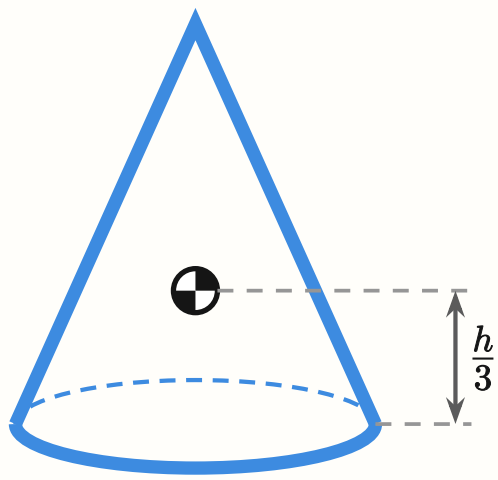

SHELL (no lid): $\frac{R}{2}$

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## 2 Cones (height $h$)




SOLID: $\frac{h}{4}$ from base




HOLLOW (curved surface): $\frac{h}{3}$

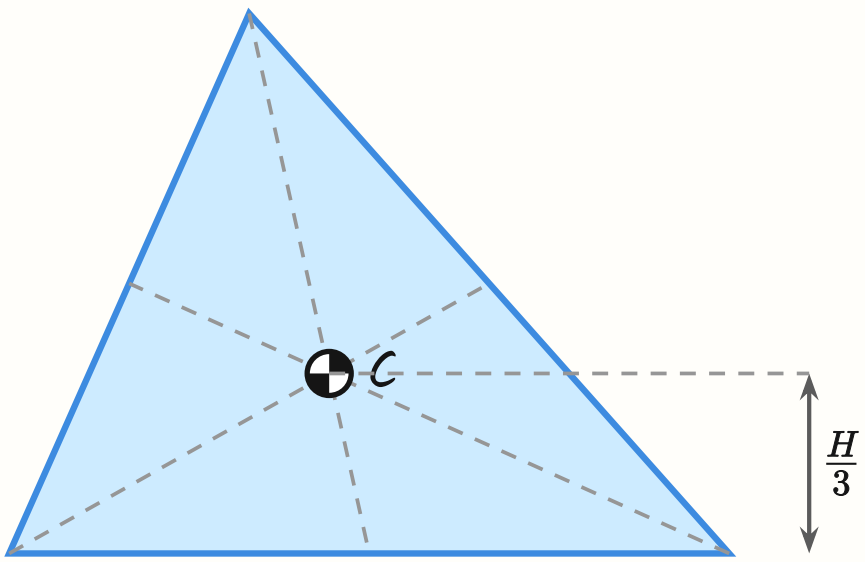
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- Solid cone: $\frac{h}{4}$ from the base $= \frac{3h}{4}$ from the apex. Hollow cone WITHOUT its base: $\frac{h}{3}$ from the base. Solid cone of $h = 24\,\text{cm}$: C is $6\,\text{cm}$ above the base.

# Triangle and Composite Bodies

*UNIFORM lamina / rods · coordinates in metres*

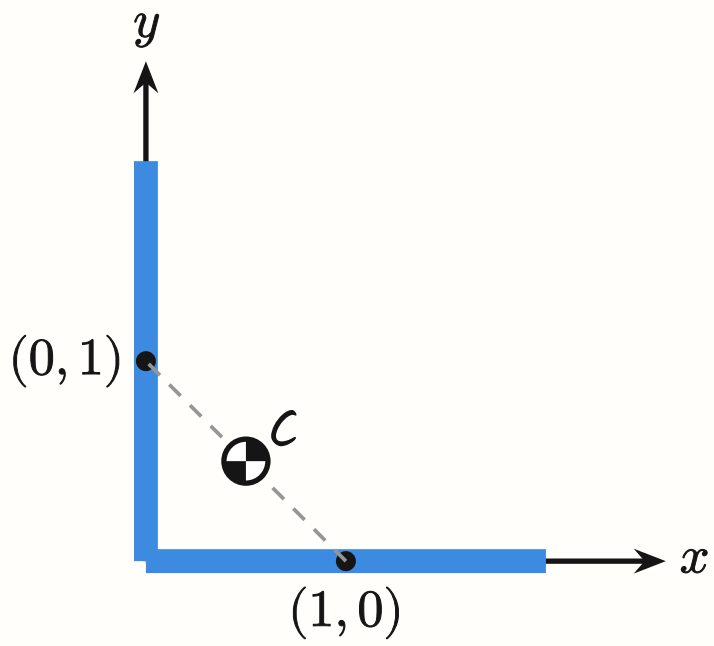
## 1 Triangular lamina




$$\vec{r}_{cm} = \frac{\vec{r}_A + \vec{r}_B + \vec{r}_C}{3}$$

*intersection of the MEDIANS (dashed) · $\frac{H}{3}$ above any side, $H$ = height on that side · lamina only (three equal rods forming a triangle: same point only if EQUILATERAL)*

## 2 Composite body: L of two rods




**Example**

*Two identical uniform rods of length $2\,\mathrm{m}$ along the axes: each acts at its middle, so C $= \left(\frac{1+0}{2}, \frac{0+1}{2}\right) = (0.5,\ 0.5)\,\mathrm{m}$, OUTSIDE the rods.*

- *Method: split the body into simple parts, replace each by its mass at its own C, then use the particle formula.*

# Cavity Method

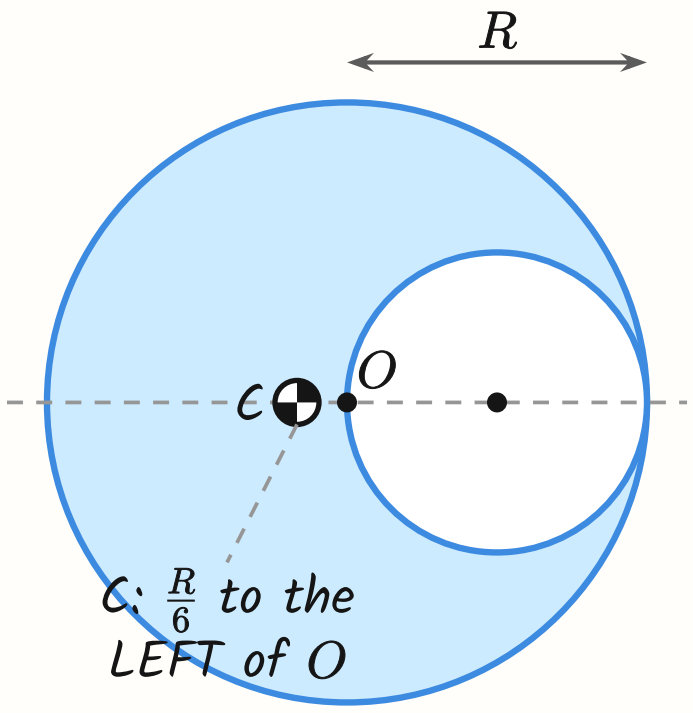
**UNIFORM bodies · removed part treated as NEGATIVE mass · $x$ from the centre $O$ of the full body**

## 1 Removed part

$$x_{cm} = \frac{m_1 x_1 - m_2 x_2}{m_1 - m_2}$$

$m_1$, $x_1$ = FULL body (before cutting) and its C · $m_2$, $x_2$ = REMOVED part and its C · uniform: masses $\propto$ areas (lamina) or volumes (solid)

## 2 Disc with a hole




**Example**

Disc of radius $R$; a hole of radius $\frac{R}{2}$ centred at $x_2 = \frac{R}{2}$: $m_2 = \frac{m_1}{4}$ (area ratio),

$x_{cm} = \frac{-\frac{m_1}{4}\cdot\frac{R}{2}}{\frac{3}{4}m_1} = -\frac{R}{6}$

C moves AWAY from the hole.

**Example**

Solid sphere of radius $R$ with a spherical cavity of radius $\frac{R}{2}$ touching its surface: $m_2 = \frac{m_1}{8}$ (volume ratio) at $\frac{R}{2}$, so $x_{cm} = -\frac{\frac{1}{8}\cdot\frac{R}{2}}{\frac{7}{8}} = -\frac{R}{14}$.

# Motion of the Centre of Mass

*system of particles · total mass $M$ CONSTANT · velocities in an INERTIAL frame*

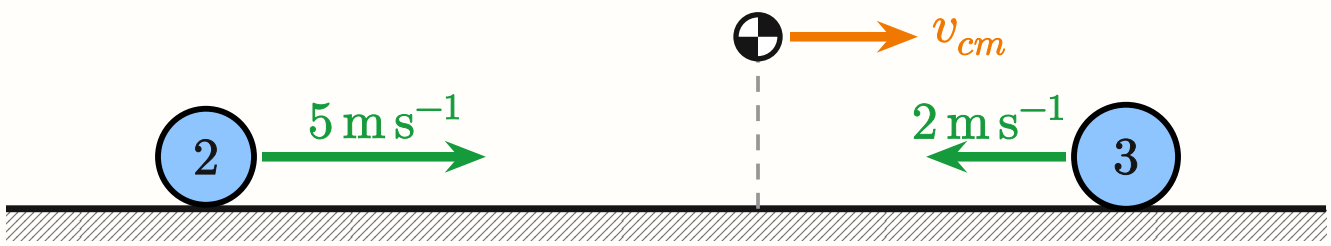
## 1 Velocity, momentum, acceleration of C

$$\vec{v}_{cm} = \frac{\sum m_i \vec{v}_i}{M} \qquad \vec{P} = M\vec{v}_{cm}$$

$$\vec{a}_{cm} = \frac{\sum m_i \vec{a}_i}{M} \qquad \vec{F}_{ext} = M\vec{a}_{cm}$$

*$\vec{P}$ = TOTAL momentum ($\mathrm{kg\,m\,s^{-1}}$) · $\vec{F}_{ext}$ = sum of EXTERNAL forces only: internal forces cancel in pairs (Newton's third law)*

## 2 Two bodies on a smooth floor




**Example**

$2\,\mathrm{kg}$ *at* $5\,\mathrm{m\,s^{-1}}$ *to the right,* $3\,\mathrm{kg}$ *at* $2\,\mathrm{m\,s^{-1}}$ *to the left:* $v_{cm} = \frac{2\times5-3\times2}{5} = 0.8\,\mathrm{m\,s^{-1}}$ *to the RIGHT.*

- *C moves as if the whole mass were at C and every external force acted there; internal forces never change $\vec{v}_{cm}$.*
- *A shell bursting in mid-air: C keeps following the ORIGINAL parabola, until a fragment hits the ground (then the ground's force is external).*

# Centre of Mass at Rest

*no EXTERNAL force (or its component) · SMOOTH ice · inertial frame*

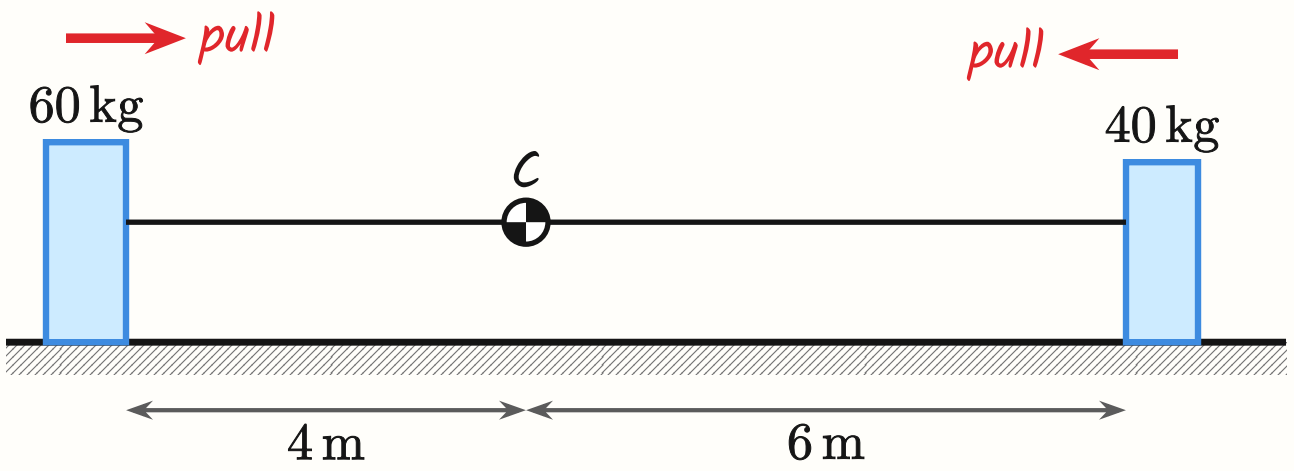
## 1 No external force

$$\vec{F}_{ext} = 0 \;\Rightarrow\; \vec{v}_{cm} = \text{constant}$$

C at rest initially: $\sum m_i \, \Delta\vec{x}_i = 0$

$\Delta\vec{x}_i$ = displacement of part $i$ (ground frame) · component form: if only $F_{ext,x} = 0$, only $v_{cm,x}$ stays constant (gravity is vertical, so the HORIZONTAL motion of C is uniform)

## 2 Two skaters pull a rope




**Example**

Skaters of $60\,\text{kg}$ and $40\,\text{kg}$, $10\,\text{m}$ apart, pull the rope: they MEET at C, $4\,\text{m}$ from the $60\,\text{kg}$ skater ($60 \times 4 = 40 \times 6$), whoever pulls.

- Internal forces (tension, muscles, explosions) change the velocities of the parts and the total KE, but NEVER $\vec{v}_{cm}$.

# Linear Momentum and Impulse

*inertial frame · $\vec{F}$ = NET force on the body · velocities along $+x$ signed*

## 1 Momentum and Newton's second law

$$\vec{p} = m\vec{v} \qquad \vec{F} = \frac{d\vec{p}}{dt}$$

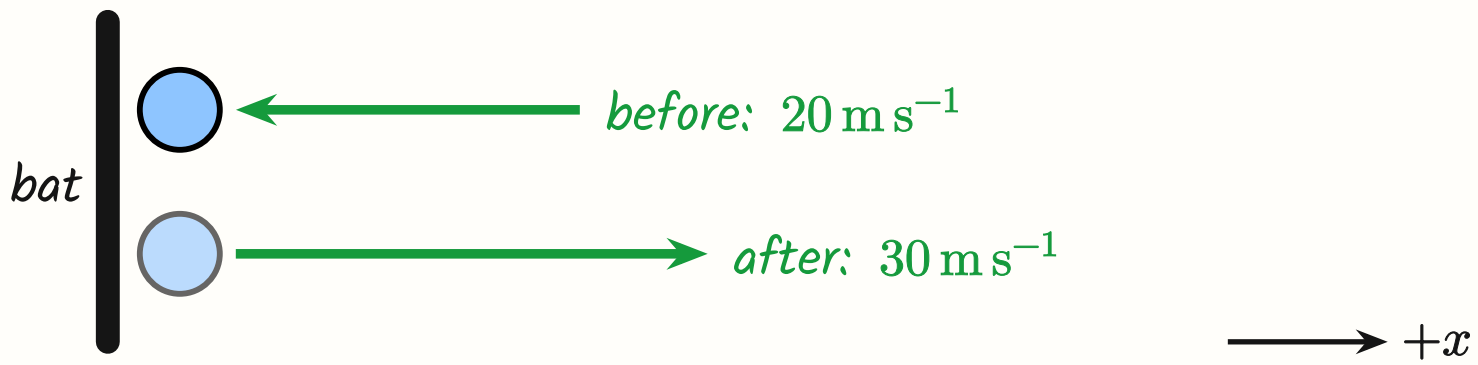
$\vec{p}$ *in* $\mathrm{kg\,m\,s^{-1}}$ · $\vec{F} = m\vec{a}$ *ONLY when the mass is constant* · $\vec{p}$ *and* $\vec{v}$ *in the same direction*

- $p = \sqrt{2mK}$ ($K$ = *kinetic energy*): *for the SAME* $K$ *the heavier body has the larger momentum; for the SAME* $p$ *the lighter body has the larger* $K$.

## 2 Impulse–momentum theorem

$$\vec{J} = \int \vec{F}\,dt = \Delta\vec{p} = \vec{p}_f - \vec{p}_i \qquad \vec{F}_{av} = \frac{\Delta\vec{p}}{\Delta t}$$

$\vec{J}$ *= impulse* ($\mathrm{N\,s} = \mathrm{kg\,m\,s^{-1}}$) *along* $\Delta\vec{p}$, *NOT along* $\vec{p}$ · $\Delta t$ *= contact time*



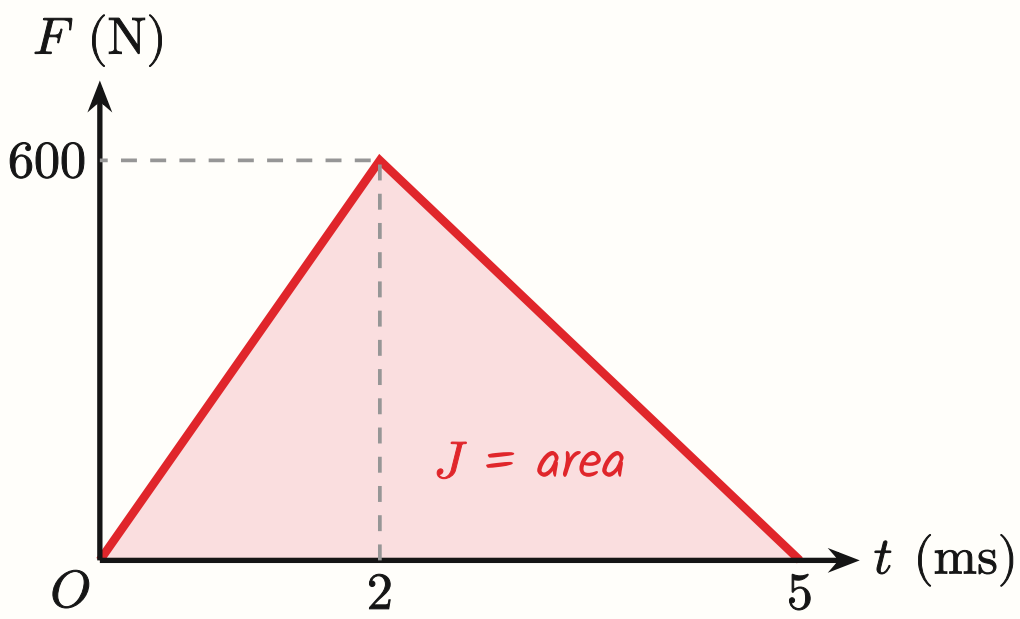

**Example**

*A* $0.15\,\mathrm{kg}$ *ball at* $20\,\mathrm{m\,s^{-1}}$ *is hit straight back at* $30\,\mathrm{m\,s^{-1}}$ *in* $0.01\,\mathrm{s}$:
$J = 0.15\,(30 - (-20)) = 7.5\,\mathrm{N\,s}$, $F_{av} = \frac{7.5}{0.01} = 750\,\mathrm{N}$ *along* $+x$.

# Force–Time Graph

*one-dimensional force · area BELOW the $t$ axis counts NEGATIVE*

## 1 Impulse = area under the $F$–$t$ graph




$$J = \int_0^t F\,dt$$

*triangle area:*

$J = \frac{1}{2} \times 600 \times 5 \times 10^{-3} = 1.5\,\text{N s}$

*· constant $F$:* $J = F\Delta t$

**Example**

*This $F$ acts on a $0.05\,\text{kg}$ ball at rest:*

$v = \frac{J}{m} = \frac{1.5}{0.05} = 30\,\text{m s}^{-1}$.

## 2 Impulsive forces

- *IMPULSIVE force: very large force for a very short time (bat, collision, explosion, a string jerking taut). Only its impulse is measurable.*
- *During the SHORT contact, finite forces (gravity, spring force) give a negligible impulse $F\Delta t$: ignore them DURING the impact only, not before or after it.*
- *Same $\Delta p$ over a LONGER time means a SMALLER average force: airbags, sand pits, bending the knees when landing.*
- *The PEAK force is not $\frac{\Delta p}{\Delta t}$: that is the AVERAGE; here $F_{av} = \frac{1.5}{0.005} = 300\,\text{N}$, half the $600\,\text{N}$ peak.*

# Conservation of Momentum

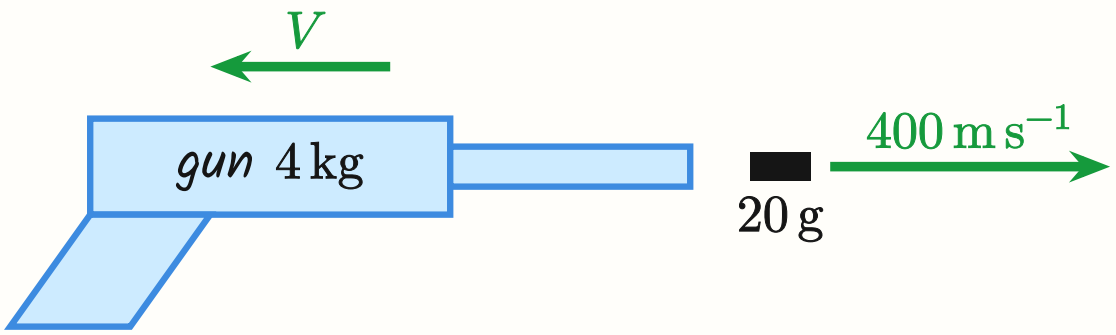
*system = all interacting bodies · inertial frame*

## 1 The law and its condition

$$\sum \vec{F}_{ext} = 0 \;\Rightarrow\; \vec{P} = \sum m_i \vec{v}_i = \text{constant}$$

*also component-wise (e.g. only horizontal when the external forces are vertical) · holds even when kinetic energy is NOT conserved*

## 2 Recoil of a gun




**Example**

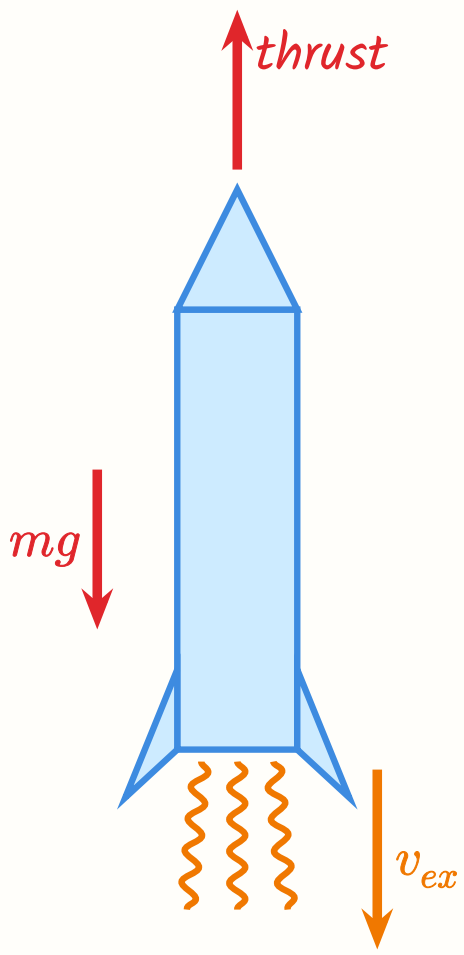
*Gun ($4\,\mathrm{kg}$) and bullet ($20\,\mathrm{g}$) at rest: $0 = 0.020 \times 400 + 4V$, so $V = -2\,\mathrm{m\,s^{-1}}$: the gun RECOILS at $2\,\mathrm{m\,s^{-1}}$.*

- *Short collisions and explosions: $\vec{P}$ just before = just after, even with gravity or friction present, as long as those forces are FINITE (impulsive approximation).*
- *Body at REST splits into two: equal and opposite momenta, so $\frac{K_1}{K_2} = \frac{m_2}{m_1}$ (the LIGHTER piece carries more KE).*
- *NOT conserved when an impulsive EXTERNAL force acts: a fixed pivot or wall, the floor under a bouncing ball, a string fixed to a support.*

# Variable Mass: Rocket

$v_{ex}$ = speed of the exhaust RELATIVE to the rocket, CONSTANT · $m_0$ = initial, $m$ = current mass

## 1 Thrust




$$F_{th} = -v_{ex}\frac{dm}{dt}$$

$\frac{dm}{dt} < 0$ (rocket loses mass), so $F_{th} > 0$, opposite to the exhaust

$$m\frac{dv}{dt} = -v_{ex}\frac{dm}{dt} - mg$$

vertical flight, uniform $g$ · LIFT-OFF needs $F_{th} > m_0 g$

## 2 Speed gained

$$\Delta v = v_{ex}\ln\frac{m_0}{m} \qquad v = v_{ex}\ln\frac{m_0}{m} - gt$$

first: GRAVITY-FREE space (no drag) · second: vertical, from REST, uniform $g$, burning for time $t$ · ln = natural log

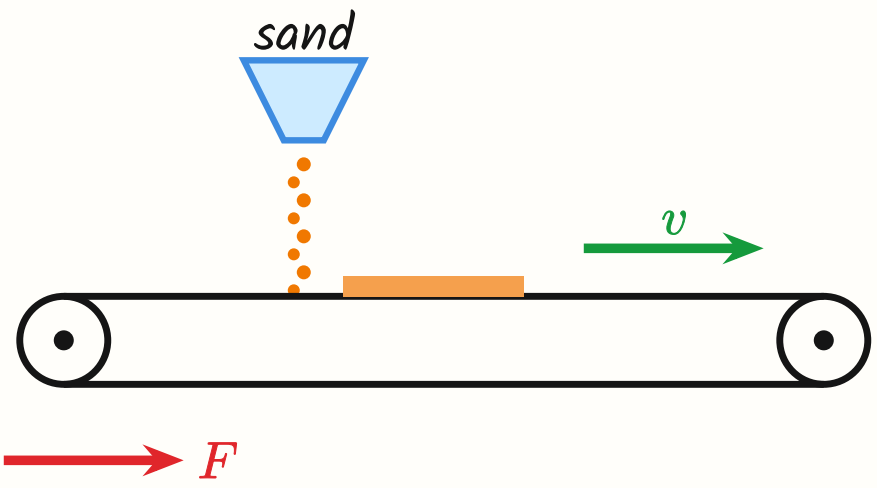
**Example**

$v_{ex} = 2000\,\mathrm{m\,s^{-1}}$, fuel burnt at $5\,\mathrm{kg\,s^{-1}}$: $F_{th} = 10^4\,\mathrm{N}$. In free space, burning two-thirds of the mass ($m_0 = 3m$):
$\Delta v = 2000\ln 3 = 2197\,\mathrm{m\,s^{-1}}$.

# Conveyor Belt and Chain

*belt speed $v$ CONSTANT · sand falls VERTICALLY · uniform flexible chain, $\lambda$ = mass per unit length*

## 1 Sand on a moving belt



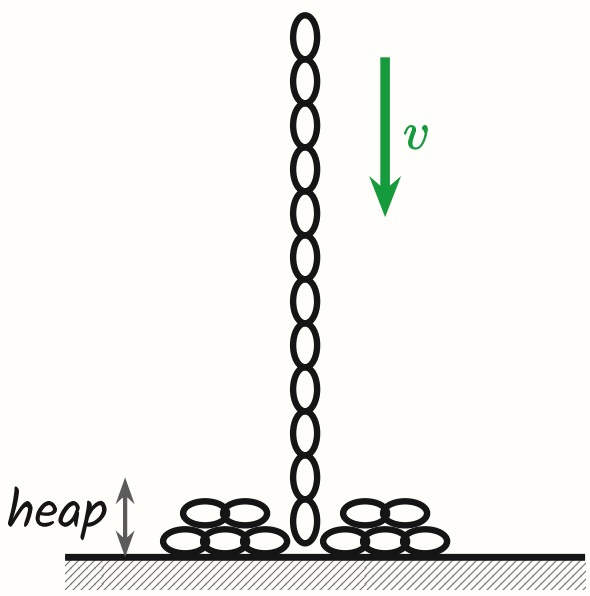

$$F = v\frac{dm}{dt}$$

*extra force to keep $v$ CONSTANT · power supplied $= Fv = v^2\frac{dm}{dt}$, but KE grows only at $\frac{1}{2}v^2\frac{dm}{dt}$; HALF becomes heat (sand slips on the belt)*

**Example**

$v = 2\,\mathrm{m\,s^{-1}}$, $\frac{dm}{dt} = 5\,\mathrm{kg\,s^{-1}}$: $F = 10\,\mathrm{N}$, power $20\,\mathrm{W}$, KE rate $10\,\mathrm{W}$, heat $10\,\mathrm{W}$.

## 2 Chain falling on a table




$$N = \lambda g\ell + \lambda v^2 = 3\lambda g\ell$$

*chain held VERTICAL, lower end touching the table, released from rest · $\ell$ = length already on the table, $v^2 = 2g\ell$ (free fall) · each link STOPS on landing · last link landing: $N = 3\times$ weight*

- *Chain LIFTED at constant speed $v$ from a heap: $F = \lambda g\ell + \lambda v^2$, $\ell$ = hanging length ($\lambda v^2$ = rate of giving momentum to new links).*

# Collisions: Types

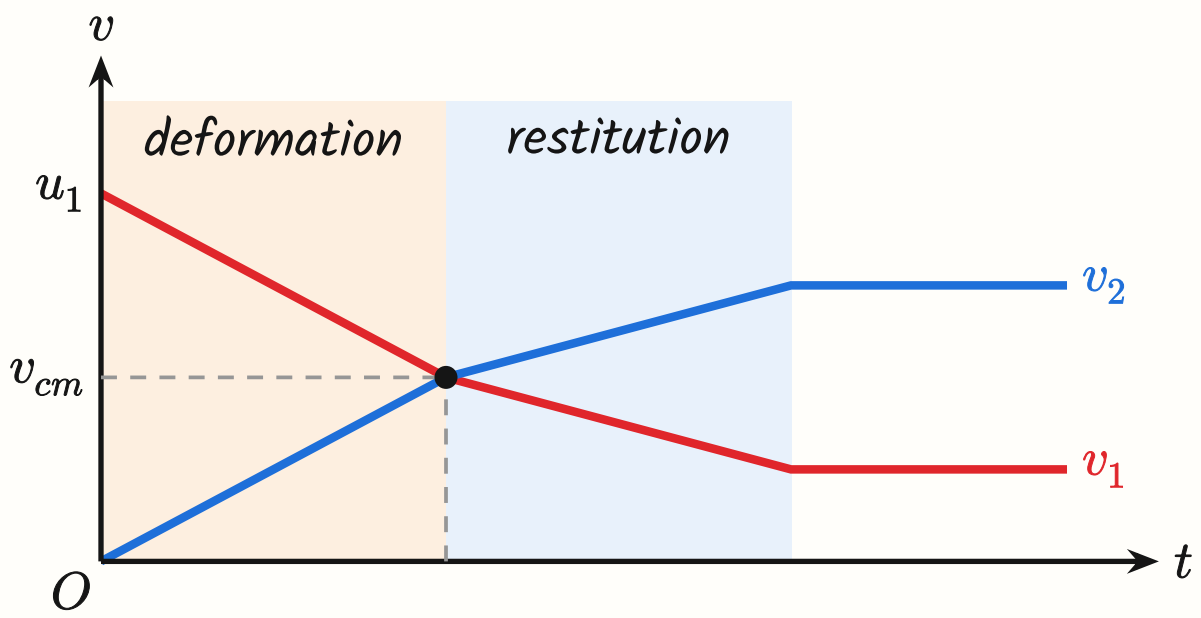
*two bodies · NO EXTERNAL impulse during the short contact*

## 1 What is conserved

| Type | momentum | kinetic energy | $e$ |
|---|---|---|---|
| ELASTIC | conserved | conserved | $1$ |
| INELASTIC | conserved | decreases | $0 < e < 1$ |
| PERFECTLY inelastic | conserved | maximum loss, bodies STICK | $0$ |

- TOTAL energy is ALWAYS conserved: the lost KE becomes heat, sound and deformation.
- Nearly elastic: hard steel or ivory balls, collisions of atoms and nuclei (no excitation). Perfectly inelastic: a bullet that EMBEDS, putty, two trolleys that couple.

## 2 Velocities during the contact




EQUAL masses, $e = \frac{1}{2}$, body 2 starts at rest. At MAXIMUM deformation both move with the SAME velocity ($= v_{cm}$); then they push apart until contact ends.

# Coefficient of Restitution

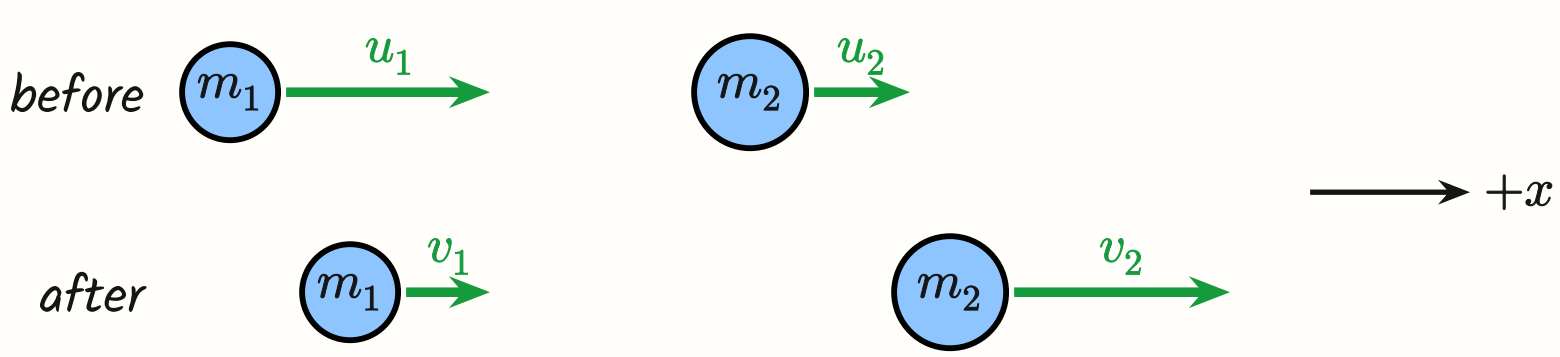
velocities ALONG the line of impact, signed, $+x$ to the right · $u$ before, $v$ after

## 1 Definition

$$e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{\text{speed of separation}}{\text{speed of approach}} = \frac{J_R}{J_D}$$

$J_D$, $J_R$ = impulses in the deformation and restitution phases · $0 \le e \le 1$ · depends on BOTH materials (and slightly on the impact speed)

## 2 Before and after




Body 1 on the LEFT: approach needs $u_1 > u_2$. Every velocity carries its sign; a body moving to the LEFT has a negative $v$.

## 3 Measuring $e$

$$e = \sqrt{\frac{h'}{h}}$$

ball DROPPED from height $h$ on a FIXED floor rebounds to $h'$ (no air drag) · $2.0\,\text{m}$ to $0.72\,\text{m}$: $e = \sqrt{0.36} = 0.6$

# 1-D Elastic Collision

HEAD-ON · perfectly ELASTIC ($e = 1$) · no external impulse · velocities signed along $+x$

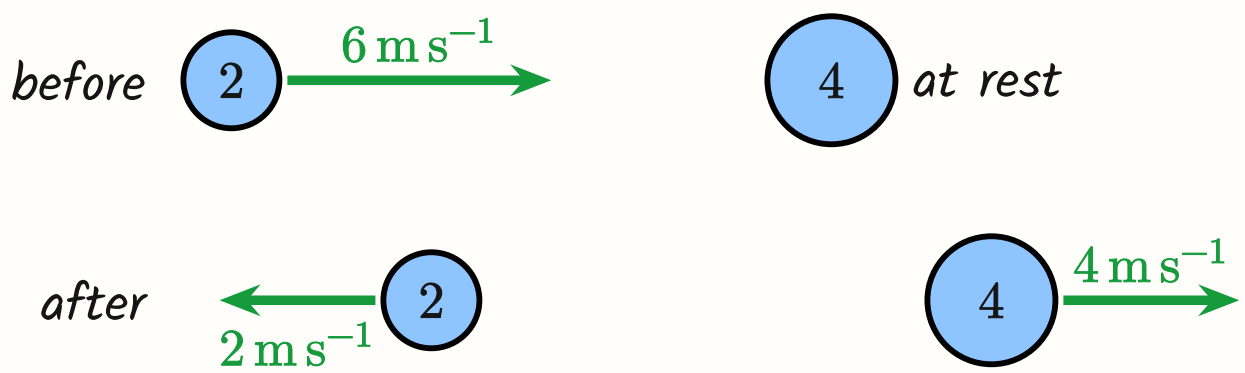
## 1 Final velocities

$$v_1 = \frac{(m_1 - m_2)u_1 + 2m_2u_2}{M}$$

$$v_2 = \frac{(m_2 - m_1)u_2 + 2m_1u_1}{M}$$

$M = m_1 + m_2$ · from momentum + kinetic energy conservation, which together give $v_2 - v_1 = u_1 - u_2$ (relative velocity reverses)

## 2 Example




**Example**

$m_1 = 2\,\text{kg}$ at $6\,\text{m s}^{-1}$ hits $m_2 = 4\,\text{kg}$ at rest: $v_1 = \frac{(2-4)\times 6}{6} = -2\,\text{m s}^{-1}$ (rebounds), $v_2 = \frac{2\times 2\times 6}{6} = 4\,\text{m s}^{-1}$. Check: KE $36\,\text{J} = 4\,\text{J} + 32\,\text{J}$.

- Shortcut: use momentum together with $v_2 - v_1 = u_1 - u_2$ (linear) instead of the KE equation (quadratic).

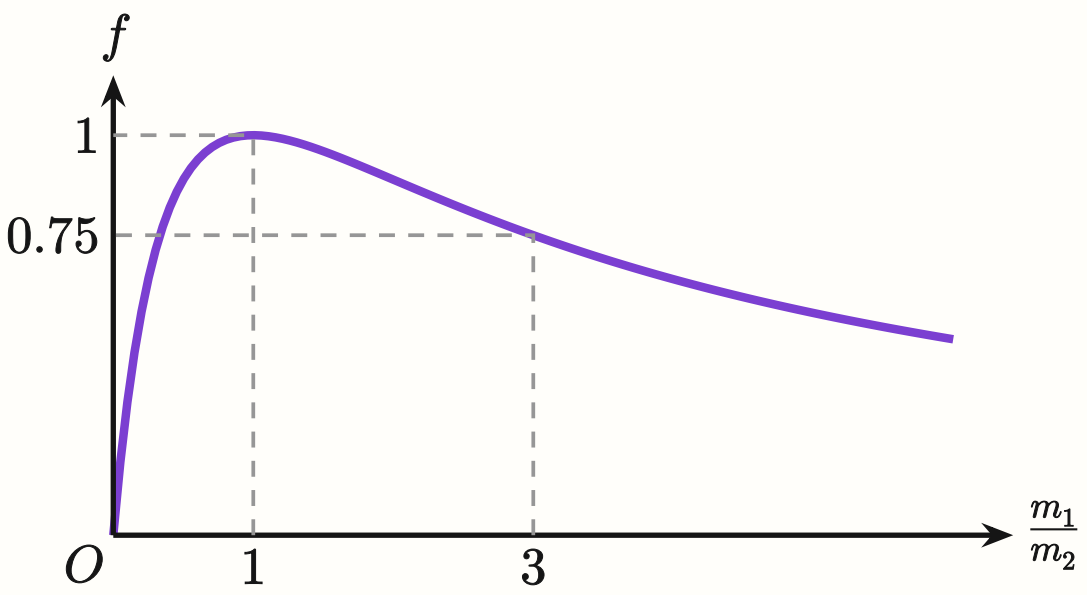
# Elastic Collision: Special Cases

**HEAD-ON · ELASTIC · $m_2$ initially AT REST unless stated**

## 1 Limits

- EQUAL masses: they EXCHANGE velocities ($v_1 = u_2$, $v_2 = u_1$), even if both move.
- $m_1 \gg m_2$: $v_1 \approx u_1$, $v_2 \approx 2u_1$ (heavy body hardly slows).
- $m_1 \ll m_2$: $v_1 \approx -u_1$, $v_2 \approx 0$ (ball rebounds from a wall with the same speed).
- $m_2$ at rest: $v_1$ has the sign of $(m_1 - m_2)$: $m_1$ REBOUNDS only if $m_1 < m_2$, STOPS if $m_1 = m_2$.

## 2 Fraction of KE transferred




$$f = \frac{4m_1m_2}{(m_1 + m_2)^2}$$

$f = 1$ only for EQUAL masses

· same $f$ for $\frac{m_1}{m_2}$ and $\frac{m_2}{m_1}$

**Example**

Neutron ($m_1$) on a carbon nucleus ($m_2 = 12m_1$) at rest: $f = \frac{4\times12}{13^2} = \frac{48}{169} = 0.284$. Moderators use LIGHT nuclei to slow neutrons.

# Collisions with Restitution $e$

**HEAD-ON · any $e$ · no external impulse · $v_{rel} = u_1 - u_2$ = relative speed of APPROACH**

## 1 Final velocities

$$v_1 = \frac{(m_1 - em_2)u_1 + (1+e)m_2u_2}{M}$$

$$v_2 = \frac{(m_2 - em_1)u_2 + (1+e)m_1u_1}{M}$$

*$e = 1$ gives the elastic result · $e = 0$: common velocity $v = \frac{m_1u_1 + m_2u_2}{M}$ (bodies STICK)*

- *$e = 0$ does NOT mean all KE is lost: the bodies keep moving together with $v_{cm}$, so $\frac{1}{2}Mv_{cm}^2$ always remains.*

## 2 Loss of kinetic energy

$$\Delta K = \frac{1}{2}\mu v_{rel}^2\,(1 - e^2) \qquad \mu = \frac{m_1m_2}{M}$$

*$\mu$ = REDUCED mass (kg) · $e = 1$: no loss · $e = 0$: MAXIMUM loss $\frac{1}{2}\mu v_{rel}^2$ (all the KE relative to C)*

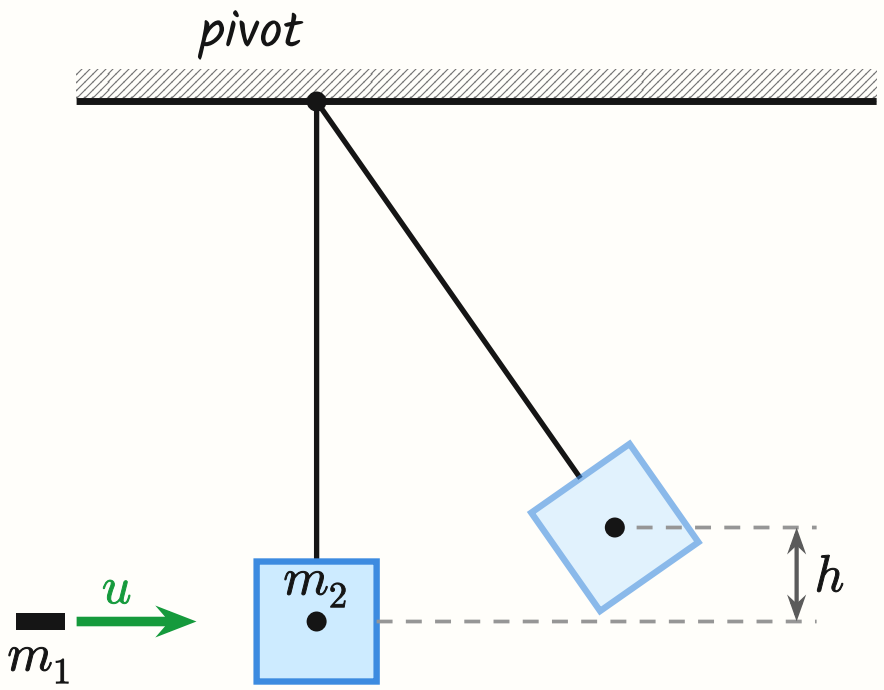
**Example**

$3\,\text{kg}$ *at* $4\,\text{m s}^{-1}$ *hits* $1\,\text{kg}$ *at rest,* $e = 0.5$*:*
$v_1 = \frac{(3-0.5)\times 4}{4} = 2.5\,\text{m s}^{-1}$, $v_2 = \frac{1.5\times 3\times 4}{4} = 4.5\,\text{m s}^{-1}$; $\mu = 0.75\,\text{kg}$,
$\Delta K = \frac{1}{2}\times 0.75\times 16\times 0.75 = 4.5\,\text{J}$ $(24\,\text{J} \to 19.5\,\text{J})$.

# Ballistic Pendulum

**bullet $m_1$ EMBEDS ($e = 0$) · block $m_2$ on a light inextensible string · uniform $g$**

## 1 Set-up




$$m_1 u = MV, \quad V = \sqrt{2gh}$$

$$u = \frac{M}{m_1}\sqrt{2gh}$$

$M = m_1 + m_2$ · impact so SHORT that the string stays vertical during it (no horizontal impulse) · $h \le$ string length (string stays taut)

## 2 Energy

$$\frac{K_{lost}}{K_{bullet}} = \frac{m_2}{M}$$

lost IN the collision (heat, deformation) · mechanical energy is conserved only in the SWING after it

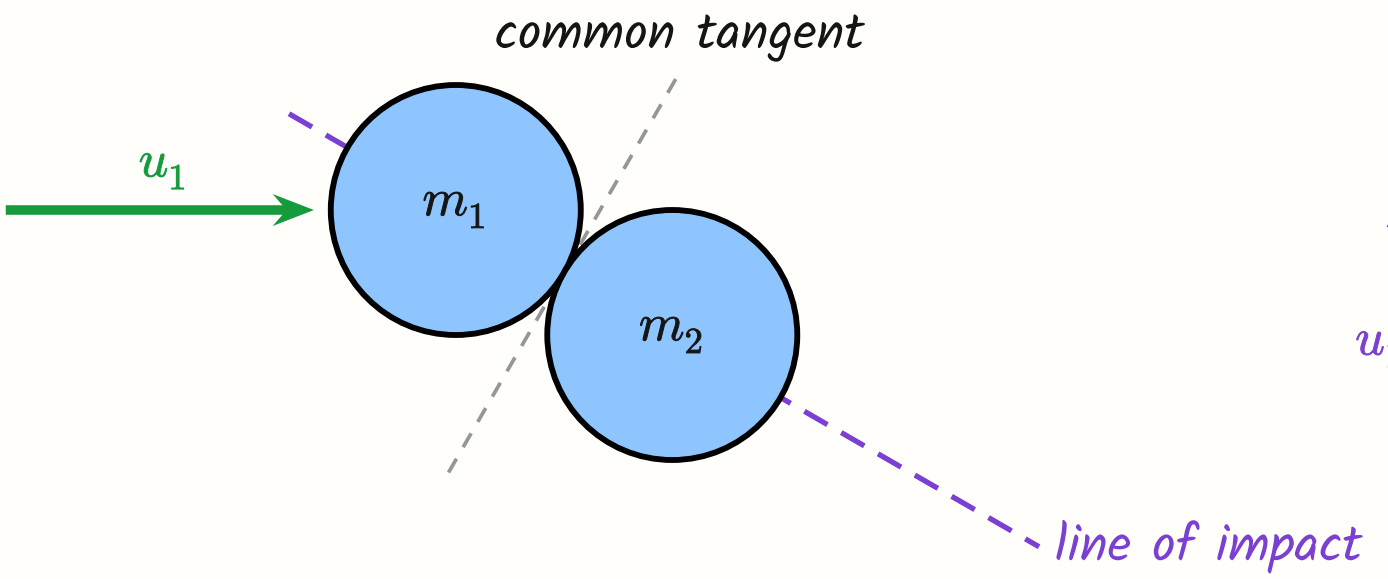
**Example**

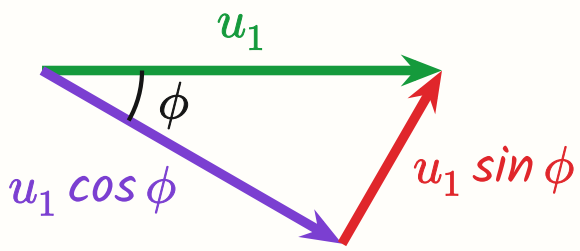
$m_1 = 10\,\mathrm{g}$, $m_2 = 0.99\,\mathrm{kg}$, $h = 5.0\,\mathrm{cm}$, $g = 10\,\mathrm{m\,s^{-2}}$;
$V = \sqrt{2 \times 10 \times 0.05} = 1.0\,\mathrm{m\,s^{-1}}$, $u = 100 \times 1.0 = 100\,\mathrm{m\,s^{-1}}$; $99\,\%$ of the bullet's KE becomes heat.

# Oblique Collisions

*SMOOTH spheres (no friction impulse along the tangent) · no external impulse · $n$ = along the line of impact, $t$ = along the tangent*

## 1 Line of impact




$\phi$ = angle between $\vec{u}_1$ and the line of impact (the line joining the CENTRES at contact). Purple: component along it; red: along the tangent.

## 2 Rules

$$m_1 u_{1n} + m_2 u_{2n} = m_1 v_{1n} + m_2 v_{2n}$$

$$v_{2n} - v_{1n} = e\,(u_{1n} - u_{2n})$$

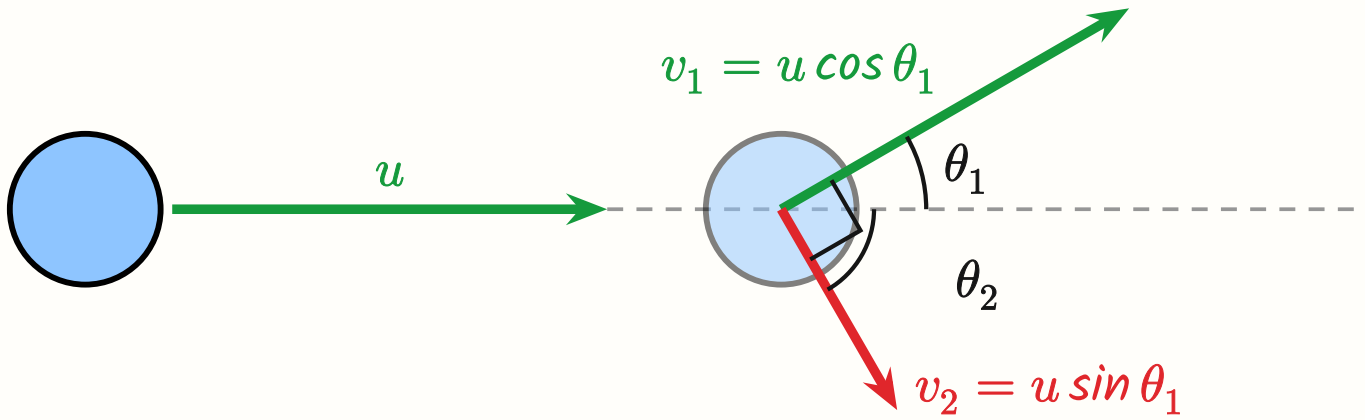
$$v_{1t} = u_{1t}, \qquad v_{2t} = u_{2t}$$

*along the line of impact: an ordinary 1-D collision with $e$ · along the tangent: each body KEEPS its own component (smooth surfaces only) · $m_2$ at rest: it moves off ALONG the line of impact*

# Equal Masses at $90°$

EQUAL masses · perfectly ELASTIC · target AT REST · NOT head-on · smooth spheres

## 1 They fly apart at a right angle




## 2 Why

$$\theta_1 + \theta_2 = 90°$$

*momentum:* $\vec{u} = \vec{v}_1 + \vec{v}_2$ · *kinetic energy:* $u^2 = v_1^2 + v_2^2$ · *both together (Pythagoras)* $\Rightarrow \vec{v}_1 \perp \vec{v}_2$

**Example**

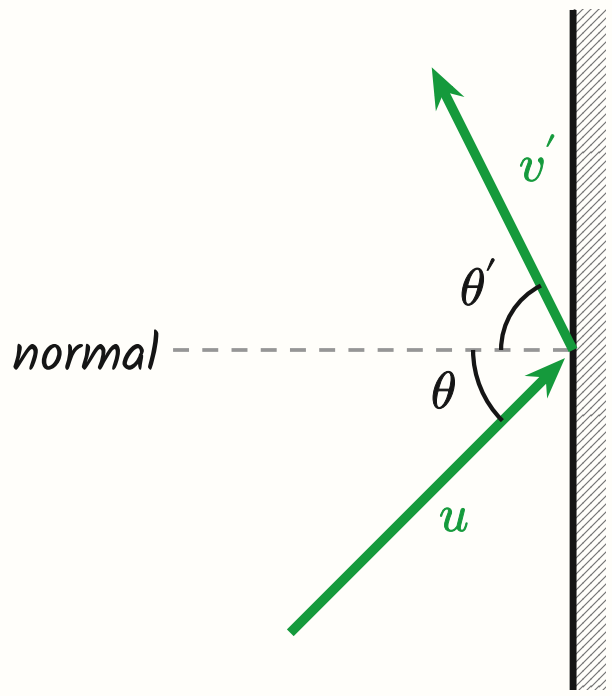
$u = 10\,\mathrm{m\,s^{-1}}$, *ball 1 deflected by* $\theta_1 = 30°$*:*
$v_1 = 10\cos 30° = 8.66\,\mathrm{m\,s^{-1}}$, $v_2 = 10\sin 30° = 5.0\,\mathrm{m\,s^{-1}}$ *at* $\theta_2 = 60°$ *on the other side.*

- *Head-on: no angle, they simply exchange velocities. Equal masses but INELASTIC: the angle between them is LESS than* $90°$.
- *Unequal masses (elastic, target at rest): angle LESS than* $90°$ *if* $m_1 > m_2$, *MORE than* $90°$ *if* $m_1 < m_2$.

# Collision with a Wall

FIXED (infinitely heavy) SMOOTH wall · angles measured from the NORMAL · restitution $e$

## 1 Oblique impact




$$\tan\theta' = \frac{\tan\theta}{e}$$

$$v' = u\sqrt{\sin^2\theta + e^2\cos^2\theta}$$

normal component: $u\cos\theta \to e\,u\cos\theta$ (reversed) · along the wall: $u\sin\theta$ unchanged (SMOOTH) · $e < 1$: rebounds FLATTER ($\theta' > \theta$)

## 2 Impulse from the wall

$$J = m(1+e)\,u\cos\theta$$

along the normal, AWAY from the wall · elastic ($e = 1$): $J = 2mu\cos\theta$, $\theta' = \theta$, speed unchanged

**Example**

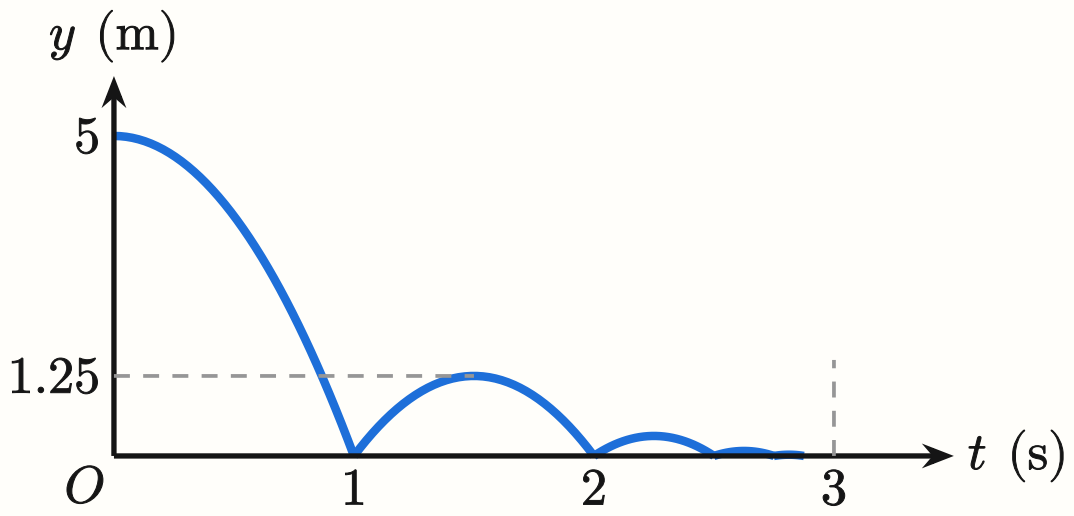
$u = 10\,\mathrm{m\,s^{-1}}$ at $\theta = 45°$, $e = 0.5$: normal $7.07 \to 3.54\,\mathrm{m\,s^{-1}}$, along the wall $7.07\,\mathrm{m\,s^{-1}}$; $v' = 10\sqrt{0.5 + 0.125} = 7.91\,\mathrm{m\,s^{-1}}$, $\tan\theta' = 2$, $\theta' = 63.4°$.

- Momentum of the BALL is not conserved: the wall (Earth) takes the impulse. Momentum along the wall IS conserved (smooth wall).

# Bouncing Ball

ball DROPPED from height $h$ on a FIXED floor · same $e$ at every bounce · no air drag · uniform $g$

## 1 Height–time graph




$h = 5\,\mathrm{m}$, $e = 0.5$, $g = 10\,\mathrm{m\,s^{-2}}$; $y$ = height of the ball above the floor. Each flight is a parabola; the flights get shorter by the factor $e$, and all of them end at $t = 3\,\mathrm{s}$.

## 2 Results

$$v_n = e^n\sqrt{2gh}, \qquad h_n = e^{2n}h$$

$$t_{tot} = \sqrt{\frac{2h}{g}}\,\frac{1+e}{1-e}, \qquad s_{tot} = h\,\frac{1+e^2}{1-e^2}$$

$v_n$, $h_n$ = rebound speed and height after the $n$-th bounce · $t_{tot}$, $s_{tot}$ = total time and total DISTANCE until it stops ($e < 1$) · $e = 1$: bounces for ever

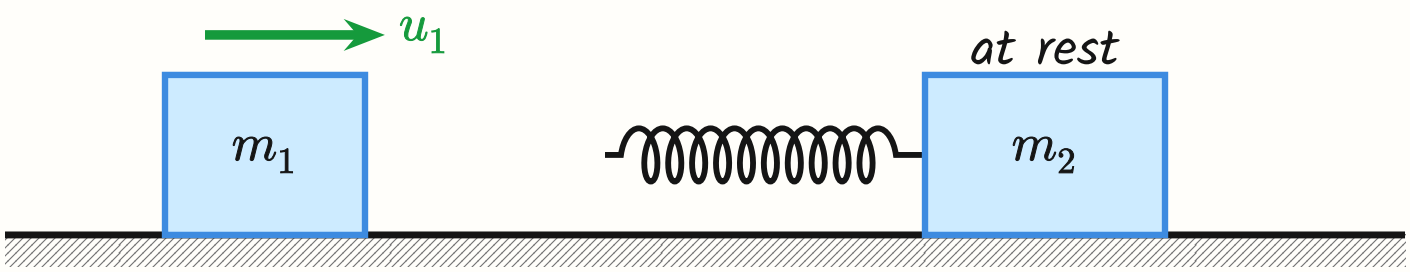
**Example**

$h = 5\,\mathrm{m}$, $e = 0.5$: $h_1 = 1.25\,\mathrm{m}$, $h_2 = 0.3125\,\mathrm{m}$; $t_{tot} = 1 \times \frac{1.5}{0.5} = 3\,\mathrm{s}$; $s_{tot} = 5 \times \frac{1.25}{0.75} = 8.33\,\mathrm{m}$.

# Two Blocks and a Spring

**SMOOTH floor · ideal MASSLESS spring of constant $k$ · $\delta$ = compression · $v_{rel} = u_1 - u_2$**

## 1 Maximum compression




$$\text{at } \delta_{max}: \quad v_1 = v_2 = v_{cm} = \frac{m_1 u_1 + m_2 u_2}{M}$$

$$\frac{1}{2}\mu v_{rel}^2 = \frac{1}{2} k \delta_{max}^2 \quad \Rightarrow \quad \delta_{max} = v_{rel}\sqrt{\frac{\mu}{k}}$$

$\mu = \frac{m_1 m_2}{M}$ · the KE relative to C is stored in the spring · when the spring regains its natural length, the result is that of an ELASTIC collision

**Example**

$m_1 = 2\,\text{kg}$ at $3\,\text{m s}^{-1}$, $m_2 = 1\,\text{kg}$ at rest, $k = 600\,\text{N m}^{-1}$:
$v_{cm} = 2\,\text{m s}^{-1}$, $\mu = \frac{2}{3}\,\text{kg}$, $\delta_{max} = 3\sqrt{\frac{2/3}{600}} = 0.1\,\text{m}$. After separation:
$v_1 = 1\,\text{m s}^{-1}$, $v_2 = 4\,\text{m s}^{-1}$.

- $m_2$ fixed (or against a wall): $\mu \to m_1$, so $\delta_{max} = u_1\sqrt{\frac{m_1}{k}}$.
- Maximum compression is when the RELATIVE velocity is zero, not when one block stops.

# Centre of Mass Frame

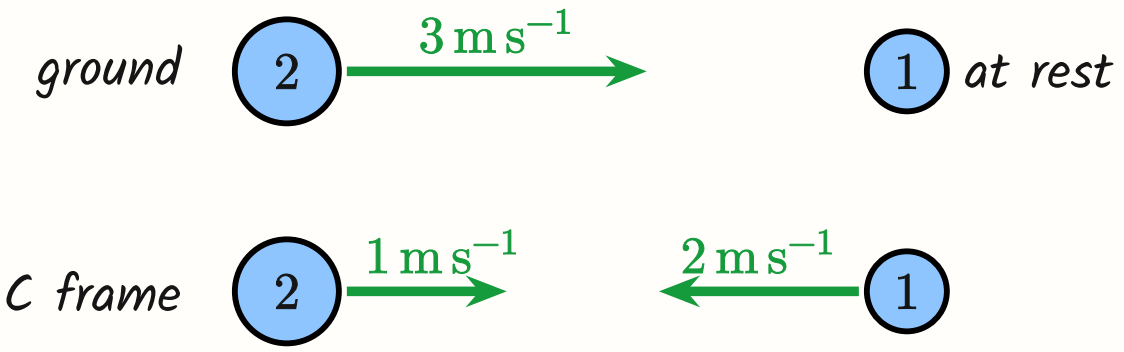
*two bodies · C frame = frame moving with $\vec{v}_{cm}$ · no external force*

## 1 Kinetic energy split

$$K = \frac{1}{2} M v_{cm}^2 + \frac{1}{2} \mu v_{rel}^2$$

$v_{rel}$ = speed of one body relative to the other, $\mu = \frac{m_1 m_2}{M}$ · first term is FIXED (no external force); ONLY the second, the KE in the C frame, can turn into heat or spring energy

## 2 Same collision, two frames




$2\,\mathrm{kg}$ and $1\,\mathrm{kg}$ bodies: $v_{cm} = 2\,\mathrm{m\,s^{-1}}$. In the C frame the momenta are $+2$ and $-2\,\mathrm{kg\,m\,s^{-1}}$: total ZERO.

**Example**

$K = 9\,\mathrm{J} = \frac{1}{2} \times 3 \times 2^2 + \frac{1}{2} \times \frac{2}{3} \times 3^2 = 6\,\mathrm{J} + 3\,\mathrm{J}$: at most $3\,\mathrm{J}$ can be lost.

- Head-on, in the C frame a collision just reverses each velocity and scales it by $e$: $v' = -e\,u'$.
- $\mu$ is smaller than either mass; $m_2 \to \infty$ (wall, Earth) gives $\mu \to m_1$.

# Explosions

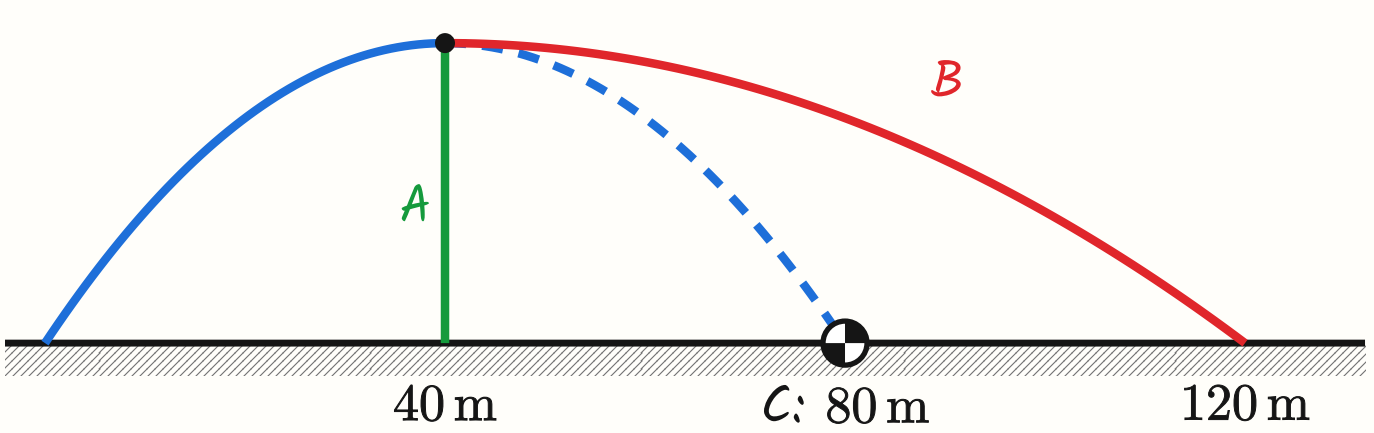
*internal IMPULSIVE forces · momentum conserved through the burst · level ground, no air drag*

## 1 Momentum, not kinetic energy

$$\vec{P}_{before} = \vec{P}_{after} \qquad K_{after} > K_{before}$$

*the burst lasts a very short time, so gravity's impulse is negligible · KE INCREASES (chemical energy released)*

## 2 Shell bursts at the top




*Range without the burst* $80\,\mathrm{m}$. *EQUAL halves at the top; A is left at REST and falls vertically; B gets TWICE the speed. Both land together, so C still lands at* $80\,\mathrm{m}$ *and B at* $120\,\mathrm{m}$.

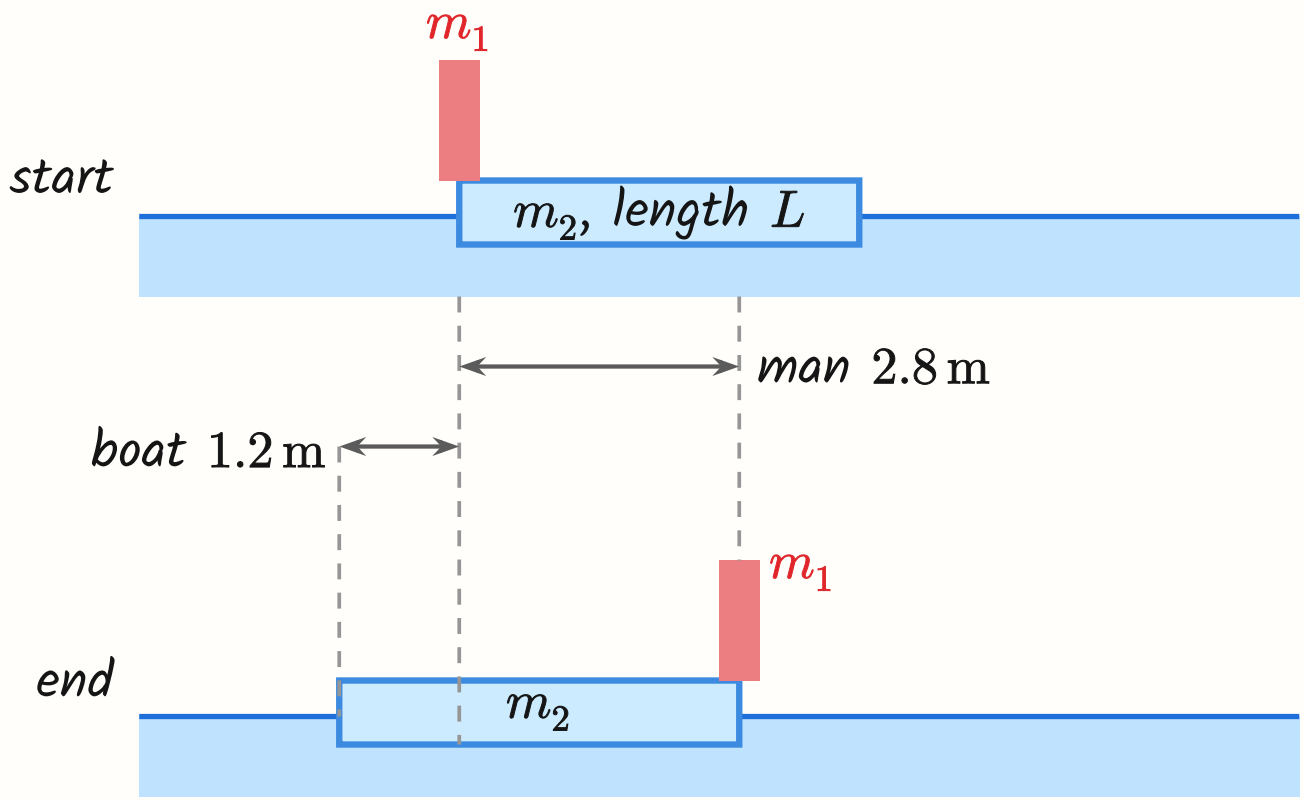
**Example**

*A shell at rest bursts:* $1\,\mathrm{kg}$ *at* $12\,\mathrm{m\,s^{-1}}$ *east,* $2\,\mathrm{kg}$ *at* $8\,\mathrm{m\,s^{-1}}$ *north, third piece* $0.5\,\mathrm{kg}$*:* $p_3 = \sqrt{12^2 + 16^2} = 20\,\mathrm{kg\,m\,s^{-1}}$, $v_3 = 40\,\mathrm{m\,s^{-1}}$, $53.1^\circ$ *south of west. Energy released* $72 + 64 + 400 = 536\,\mathrm{J}$.

# Man on a Boat

no horizontal force from the water (no drag) · all at REST initially · $m_1$ = man, $m_2$ = boat or plank

## 1 Man walks the full length $L$




## 2 Equations

$$m_1\Delta x_1 + m_2\Delta x_2 = 0, \quad \Delta x_1 - \Delta x_2 = L$$

$$\Delta x_2 = -\frac{m_1 L}{M}, \qquad \Delta x_1 = \frac{m_2 L}{M}$$

$\Delta x_1$, $\Delta x_2$ = displacements of man and boat (ground frame) · the boat moves BACKWARDS · independent of how fast or how unevenly he walks

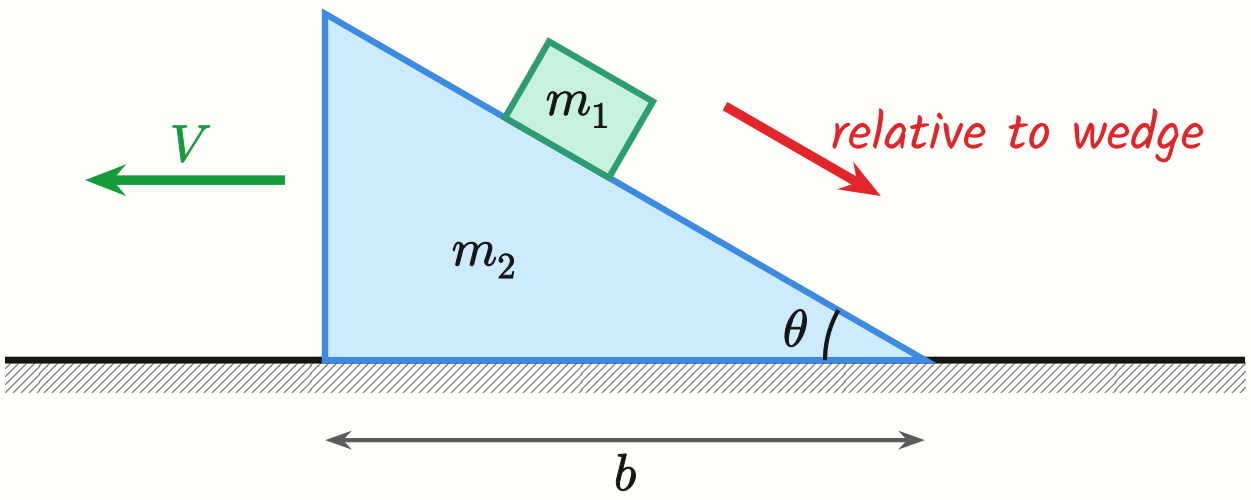
**Example**

$m_1 = 60\,\text{kg}$, $m_2 = 140\,\text{kg}$, $L = 4\,\text{m}$: boat moves $\frac{60\times 4}{200} = 1.2\,\text{m}$ back, man $2.8\,\text{m}$ forward (relative to the ground).

# Block on a Wedge

**SMOOTH incline and SMOOTH floor · block $m_1$, wedge $m_2$ · released from REST**

## 1 Set-up




## 2 What is conserved

$$m_1 v_{1x} = m_2 V \qquad \Delta x_{wedge} = \frac{m_1 b}{M}$$

*HORIZONTAL momentum only (gravity and the floor's normal force are vertical); vertical momentum is NOT conserved · $b$ = horizontal distance the block moves RELATIVE to the wedge · energy (all smooth):*

$$m_1 g h = \tfrac{1}{2} m_1 v_1^2 + \tfrac{1}{2} m_2 V^2$$

**Example**

$m_1 = 1\,\text{kg}$, $m_2 = 4\,\text{kg}$, *base* $b = 0.5\,\text{m}$: *when the block reaches the bottom the wedge has moved* $\frac{1 \times 0.5}{5} = 0.1\,\text{m}$ *back, the block* $0.4\,\text{m}$ *forward.*

- *Wedge FIXED to the floor: the floor pushes it horizontally, so horizontal momentum is NOT conserved.*

# Standard Results

## 1 One-line results

| Quantity | Result | Condition |
|---|---|---|
| Semicircular ring | $\frac{2R}{\pi}$ | uniform, from centre |
| Semicircular disc | $\frac{4R}{3\pi}$ | uniform lamina |
| Hemisphere | solid $\frac{3R}{8}$, shell $\frac{R}{2}$ | from the flat face |
| Cone | solid $\frac{h}{4}$, hollow $\frac{h}{3}$ | from the base |
| Removed part | $\frac{m_1x_1-m_2x_2}{m_1-m_2}$ | uniform density |
| Motion of C | $\vec{F}_{ext}=M\vec{a}_{cm}$ | $M$ constant |
| Impulse | $\vec{J}=\int\vec{F}dt=\Delta\vec{p}$ | any force |
| Momentum | $\vec{P}$ constant | $\sum\vec{F}_{ext}=0$ |
| Rocket | $\Delta v=v_{ex}\ln\frac{m_0}{m}$ | gravity-free |
| Restitution | $e=\frac{v_2-v_1}{u_1-u_2}$ | along line of impact |
| Elastic, at rest | $v_1=\frac{m_1-m_2}{M}u_1,\ v_2=\frac{2m_1}{M}u_1$ | head-on, $u_2=0$ |
| KE loss | $\frac{1}{2}\mu v_{rel}^2(1-e^2)$ | head-on |
| Ballistic | $u=\frac{M}{m_1}\sqrt{2gh}$ | bullet embeds |
| Equal masses | $\theta_1+\theta_2=90^\circ$ | elastic, oblique, at rest |
| Bounce | $h_n=e^{2n}h$ | fixed floor |
| Spring | $\delta_{max}=v_{rel}\sqrt{\mu/k}$ | smooth floor |
| Man on boat | $\frac{m_1L}{M}$ back | no water drag |

**Summary**

$\vec{F}_{ext}=M\vec{a}_{cm}$ • $\vec{J}=\Delta\vec{p}$ • $e=\frac{v_2-v_1}{u_1-u_2}$ • $\Delta K=\frac{1}{2}\mu v_{rel}^2(1-e^2)$

no external impulse · head-on unless stated · smooth surfaces · uniform bodies

# Common Traps

## 1 Exam traps

**Avoid these**

- ✗ "KE is conserved in a collision" (no: only in a perfectly ELASTIC one; momentum is conserved in all if no external impulse; TOTAL energy always).
- ✗ $e = 0$ means the bodies stop (no: they move TOGETHER with $v_{cm}$).
- ✗ Using $e$ along the TANGENT in an oblique collision (no: only along the line of impact).
- ✗ Momentum of a ball bouncing off the floor is conserved (no: the floor gives an external impulse).
- ✗ Semicircular ring $\frac{4R}{3\pi}$ (no: $\frac{2R}{\pi}$; $\frac{4R}{3\pi}$ is the DISC); hollow cone $\frac{h}{4}$ (no: $\frac{h}{3}$).
- ✗ $F = ma$ for a rocket or a conveyor (no: $F = \frac{dp}{dt}$ with the mass change).
- ✗ Average force from a drop on the floor without adding $mg$ to get the NORMAL force.
- ✗ Equal masses always go off at $90°$ (no: only elastic, target at rest, not head-on).
- ✗ Ballistic pendulum: energy conservation through the impact (no: only in the swing after it).

**Golden points**

- ★ Ask: EXTERNAL impulse along which direction? Conserve momentum only along the free directions; use $e$ along the line of impact; KE is conserved only when $e = 1$.

# Questions: JEE Main Level (I)

*uniform bodies · $g = 10\,\mathrm{m\,s^{-2}}$ · no air drag · fixed, smooth floor*

**Q1 · JEE Main level · MCQ**

A hole of radius $4\,\mathrm{cm}$ is cut in a uniform disc of radius $12\,\mathrm{cm}$, centred $6\,\mathrm{cm}$ from its centre. C shifts: (A) $0.75\,\mathrm{cm}$ away from the hole (B) $0.75\,\mathrm{cm}$ towards it (C) $0.67\,\mathrm{cm}$ away (D) $1.5\,\mathrm{cm}$ away

Solution: $\frac{m_2}{m_1} = \frac{4^2}{12^2} = \frac{1}{9}$; $x_{cm} = -\frac{\frac{1}{9}\times 6}{\frac{8}{9}} = -0.75\,\mathrm{cm}$.

**Answer: (A)**

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**Q2 · JEE Main level · Numerical**

A $3\,\mathrm{m}$ rod has $\lambda \propto x^2$ ($x$ from end A). Distance of C from A?

Solution: $x_{cm} = \frac{\int_0^L x^3 dx}{\int_0^L x^2 dx} = \frac{3}{4}L = \frac{3}{4}\times 3 = 2.25\,\mathrm{m}$.

**Answer:** $2.25\,\mathrm{m}$

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**Q3 · JEE Main level · MCQ**

A $0.2\,\mathrm{kg}$ ball dropped from $5\,\mathrm{m}$ rebounds to $1.8\,\mathrm{m}$; contact $0.02\,\mathrm{s}$. Average NORMAL force: (A) 160 (B) 162 (C) 40 (D) $2\,\mathrm{N}$

Solution: $10\,\mathrm{m\,s^{-1}}$ down, $6\,\mathrm{m\,s^{-1}}$ up ($\sqrt{2gh}$):
$\Delta p = 0.2 \times 16 = 3.2\,\mathrm{N\,s}$; $N - mg = \frac{3.2}{0.02} = 160$, $N = 162\,\mathrm{N}$.

**Answer: (B)**

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**Q4 · JEE Main level · Numerical**

$1\,\mathrm{kg}$ at $9\,\mathrm{m\,s^{-1}}$ hits $2\,\mathrm{kg}$ at rest, head-on, elastic. Speed of the $2\,\mathrm{kg}$ block? Fraction of KE it gets?

Solution: $v_2 = \frac{2\times 1\times 9}{3} = 6\,\mathrm{m\,s^{-1}}$; $f = \frac{4\times 1\times 2}{3^2} = \frac{8}{9} = 0.889$.

**Answer:** $6\,\mathrm{m\,s^{-1}}$, $0.889$

# Questions: JEE Main Level (2)

*head-on unless stated · $g = 10\,\mathrm{m\,s^{-2}}$ · no air drag · smooth ice*

**Q5 · JEE Main level · MCQ**

$2\,\mathrm{kg}$ at $4\,\mathrm{m\,s^{-1}}$ and $3\,\mathrm{kg}$ at $1\,\mathrm{m\,s^{-1}}$ move TOWARDS each other and stick. KE lost: (A) 15 (B) 17.5 (C) 2.5 (D) $12.5\,\mathrm{J}$

Solution: $v = \frac{2\times4-3\times1}{5} = 1\,\mathrm{m\,s^{-1}}$; $\mu = \frac{6}{5} = 1.2\,\mathrm{kg}$, $v_{rel} = 5$; $\Delta K = \frac{1}{2}\times 1.2 \times 25 = 15\,\mathrm{J}$ $(17.5 \to 2.5\,\mathrm{J})$.

**Answer: (A)**

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**Q6 · JEE Main level · Numerical**

A $6000\,\mathrm{kg}$ rocket stands on the pad; exhaust speed $1500\,\mathrm{m\,s^{-1}}$ relative to it. Minimum fuel-burn rate to lift off?

Solution: $v_{ex}\left|\frac{dm}{dt}\right| > m_0 g$; $\left|\frac{dm}{dt}\right| > \frac{6000\times10}{1500} = 40\,\mathrm{kg\,s^{-1}}$.

**Answer:** $40\,\mathrm{kg\,s^{-1}}$

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**Q7 · JEE Main level · MCQ**

A ball dropped from $3.2\,\mathrm{m}$ on a fixed floor ($e = 0.8$). Total time until it stops bouncing: (A) 0.8 (B) 1.44 (C) 7.2 (D) $6.4\,\mathrm{s}$

Solution: $\sqrt{\frac{2\times3.2}{10}} = 0.8\,\mathrm{s}$; $t_{tot} = 0.8 \times \frac{1.8}{0.2} = 7.2\,\mathrm{s}$.

**Answer: (C)**

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**Q8 · JEE Main level · Numerical**

A $50\,\mathrm{kg}$ man walks from one end of a $6\,\mathrm{m}$, $100\,\mathrm{kg}$ plank to the other; the plank lies on smooth ice. How far does the plank move?

Solution: $\Delta x_2 = \frac{m_1 L}{M} = \frac{50\times6}{150} = 2\,\mathrm{m}$, opposite to the man.

**Answer:** $2\,\mathrm{m}$

# Questions: JEE Advanced Level (I)

*SMOOTH floor · ideal massless spring · smooth spheres · no external impulse*

**A1 · JEE Advanced level · Numerical, multi-step**

A $20\,\mathrm{g}$ bullet at $200\,\mathrm{m\,s^{-1}}$ embeds in block A ($1.98\,\mathrm{kg}$). A is joined by a spring ($k = 300\,\mathrm{N\,m^{-1}}$) to block B ($2\,\mathrm{kg}$), both at rest. Find (i) A's speed just after impact, (ii) the heat produced, (iii) the maximum compression, (iv) the velocities when the spring is next at its natural length.

Solution: (i) $0.02 \times 200 = 2V$: $V = 2\,\mathrm{m\,s^{-1}}$ (spring not yet compressed). (ii) $\tfrac{1}{2} \times 0.02 \times 200^2 - \tfrac{1}{2} \times 2 \times 2^2 = 400 - 4 = 396\,\mathrm{J}$. (iii) $\mu = \tfrac{2\times2}{4} = 1\,\mathrm{kg}$, $v_{rel} = 2$: $\tfrac{1}{2} \times 1 \times 4 = \tfrac{1}{2} \times 300\,\delta^2$, $\delta = \sqrt{\tfrac{4}{300}} = 0.115\,\mathrm{m}$. (iv) Equal masses, elastic (spring): exchange, A at rest, B at $2\,\mathrm{m\,s^{-1}}$.

**Answer:** (i) $2\,\mathrm{m\,s^{-1}}$ (ii) $396\,\mathrm{J}$ (iii) $11.5\,\mathrm{cm}$ (iv) $0$, $2\,\mathrm{m\,s^{-1}}$

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**A2 · JEE Advanced level · Numerical, multi-step**

Ball 1 ($2\,\mathrm{kg}$, $5\,\mathrm{m\,s^{-1}}$) hits ball 2 ($3\,\mathrm{kg}$, at rest); at contact the line of centres makes angle $\phi$ with $\vec{u}_1$, $\cos\phi = 0.8$; $e = 0.5$. Find (i) ball 2's velocity, (ii) ball 1's speed, (iii) the KE lost, (iv) ball 1's deflection.

Solution: Along the line: $u_{1n} = 4$, tangent $u_{1t} = 3\,\mathrm{m\,s^{-1}}$ (kept). (i) $v_{2n} = \tfrac{1.5\times2\times4}{5} = 2.4\,\mathrm{m\,s^{-1}}$ along the line of centres. (ii) $v_{1n} = \tfrac{(2-1.5)\times4}{5} = 0.4$: $v_1 = \sqrt{0.4^2 + 3^2} = 3.03\,\mathrm{m\,s^{-1}}$. (iii) $\tfrac{1}{2} \times 1.2 \times 4^2 \times 0.75 = 7.2\,\mathrm{J}$ ($25 \to 17.8\,\mathrm{J}$). (iv) $\tan^{-1}\tfrac{3}{0.4} - \phi = 82.4^\circ - 36.9^\circ = 45.5^\circ$.

**Answer:** (i) $2.4\,\mathrm{m\,s^{-1}}$ (ii) $3.03\,\mathrm{m\,s^{-1}}$ (iii) $7.2\,\mathrm{J}$ (iv) $45.5^\circ$

# Questions: JEE Advanced Level (2)

$g = 10\,\mathrm{m\,s^{-2}}$ *uniform · no air drag · level ground · vertical rocket flight*

**A3 · JEE Advanced level · Numerical, multi-step**

*A* $3\,\mathrm{kg}$ *shell is fired at* $50\,\mathrm{m\,s^{-1}}$ *at angle* $\theta_0$ *above the ground* ($\sin\theta_0 = 0.8$). *At the top it splits:* $1\,\mathrm{kg}$ *retraces its path back to the gun. Find (i) the other piece's velocity, (ii) where it lands, (iii) the energy released, (iv) where C lands.*

*Solution: Top:* $t = \frac{50\times0.8}{10} = 4\,\mathrm{s}$, $x = 30\times4 = 120\,\mathrm{m}$, *speed* $30\,\mathrm{m\,s^{-1}}$. *(i)* $3\times30 = 1\times(-30) + 2v$: $v = 60\,\mathrm{m\,s^{-1}}$, *horizontal. (ii) Falls* $4\,\mathrm{s}$: $120 + 60\times4 = 360\,\mathrm{m}$. *(iii)* $\frac{1}{2}\times1\times30^2 + \frac{1}{2}\times2\times60^2 - \frac{1}{2}\times3\times30^2 = 450 + 3600 - 1350 = 2700\,\mathrm{J}$. *(iv)* $\frac{1\times0+2\times360}{3} = 240\,\mathrm{m}$ *= the original range.*

**Answer:** *(i)* $60\,\mathrm{m\,s^{-1}}$ *(ii)* $360\,\mathrm{m}$ *(iii)* $2700\,\mathrm{J}$ *(iv)* $240\,\mathrm{m}$

**A4 · JEE Advanced level · Numerical, multi-step**

*A rocket of* $1000\,\mathrm{kg}$ ($600\,\mathrm{kg}$ *fuel) burns* $10\,\mathrm{kg\,s^{-1}}$ *with* $v_{ex} = 2000\,\mathrm{m\,s^{-1}}$, *rising vertically from rest. Find (i) the thrust, (ii) the initial acceleration, (iii) the speed at burn-out, (iv) the acceleration just before burn-out.*

*Solution: (i)* $2000\times10 = 2\times10^4\,\mathrm{N}$. *(ii)* $\frac{20000-1000\times10}{1000} = 10\,\mathrm{m\,s^{-2}}$. *(iii) Burn-out at* $t = 60\,\mathrm{s}$, $m = 400\,\mathrm{kg}$: $v = 2000\ln2.5 - 10\times60 = 1833 - 600 = 1233\,\mathrm{m\,s^{-1}}$. *(iv)* $\frac{20000-400\times10}{400} = 40\,\mathrm{m\,s^{-2}}$.

**Answer:** *(i)* $2\times10^4\,\mathrm{N}$ *(ii)* $10\,\mathrm{m\,s^{-2}}$ *(iii)* $1233\,\mathrm{m\,s^{-1}}$ *(iv)* $40\,\mathrm{m\,s^{-2}}$

# Questions: JEE Advanced Level (3)

**SMOOTH incline and floor · $g = 10\,\mathrm{m\,s^{-2}}$ · smooth spheres**

**AS · JEE Advanced level · Numerical, multi-step**

A $1\,\mathrm{kg}$ block is released from rest at the top of a $2\,\mathrm{kg}$ wedge (incline $30°$, height $0.6\,\mathrm{m}$) on a smooth floor. When the block reaches the bottom, find (i) its speed $v_r$ relative to the wedge, (ii) the wedge's speed, (iii) the block's speed (ground), (iv) the wedge's displacement.

Solution: Horizontal momentum: $1 \times (v_r \cos 30° - V) = 2V$, so $V = \frac{v_r}{2\sqrt{3}}$. Energy: $1 \times 10 \times 0.6 = \frac{1}{2} \times 2V^2 + \frac{1}{2}\left[(v_r \cos 30° - V)^2 + (v_r \sin 30°)^2\right] = \left(\frac{1}{12} + \frac{7}{24}\right) v_r^2 = \frac{3}{8} v_r^2$. (i) $v_r^2 = 16$: $v_r = 4\,\mathrm{m\,s^{-1}}$. (ii) $V = \frac{4}{2\sqrt{3}} = 1.15\,\mathrm{m\,s^{-1}}$. (iii) $\sqrt{2.31^2 + 2^2} = 3.06\,\mathrm{m\,s^{-1}}$. (iv) $b = \frac{0.6}{\tan 30°} = 1.04\,\mathrm{m}$: $\frac{1 \times 1.04}{3} = 0.346\,\mathrm{m}$.

**Answer: (i) $4\,\mathrm{m\,s^{-1}}$ (ii) $1.15\,\mathrm{m\,s^{-1}}$ (iii) $3.06\,\mathrm{m\,s^{-1}}$ (iv) $0.346\,\mathrm{m}$**

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**A6 · JEE Advanced level · Multi-correct**

A smooth ball hits an IDENTICAL ball at rest, elastically and NOT head-on. Which are correct? (A) after the collision they move at $90°$ to each other (B) total KE is conserved (C) each ball's momentum is conserved (D) the struck ball moves along the line of centres at contact

Solution: (A) equal masses, elastic, target at rest: $\vec{v}_1 \perp \vec{v}_2$. (B) elastic. (C) wrong: only the TOTAL momentum is conserved. (D) the only impulse on it acts along the line of centres (smooth balls).

**Answer: (A), (B), (D)**